Scale and Scope in the Value of Data*

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Abstract

Claims of diminishing returns to data measure data by volume. I model data as three inputs: observations, their attributes, and those a firm predicts with. In high dimensions, a learning multiplier links the attributes: with each attribute it trains on, the firm learns the others' effects better, so returns to attributes can increase. The multiplier, a sufficient statistic for these spillovers, can be measured by estimating a learning curve. Learning precision grows more than proportionally with observations, and returns to them can increase, provided that the firm trains on attributes stronger than some it predicts with and samples are small.

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1 Introduction

Do returns to data diminish? The answer shapes how firms that predict compete: if returns diminish, a firm that holds much data gains little over its rivals from more of it; if they increase, data favor the firms that already hold the most, a concern of competition authorities.[1] The evidence points both ways. Demand forecasts for a product improve at a diminishing rate as

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[1]See Subcommittee on Antitrust, Commercial, and Administrative Law of the Committee on the Judiciary (2022), Stigler Committee on Digital Platforms (2019), Furman et al. (2019), and the EU Data Act (European Parliament and Council of the European Union 2023).

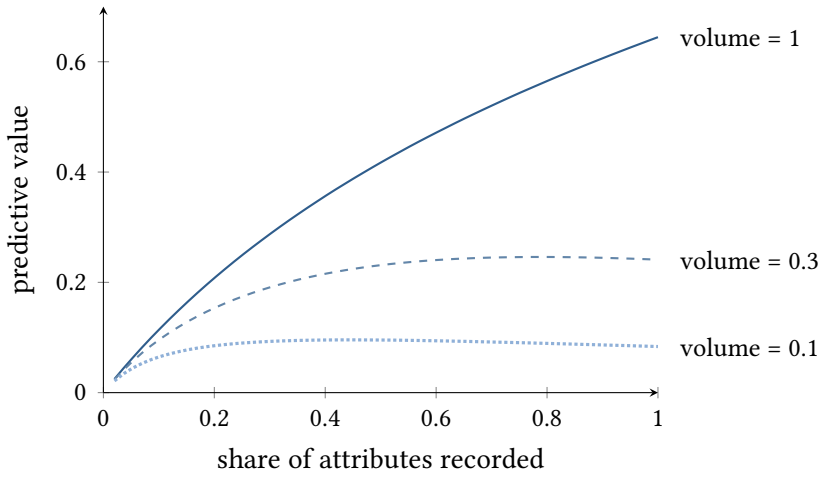


Figure 1. Datasets of equal volume. Each curve holds fixed the volume of a dataset, the number of entries it records, and varies how the volume splits between individuals and the attributes recorded for each; labels give the volume relative to the largest. The firm predicts with every attribute it records. Value is the share of the R^2 of a firm that knows every weight and observes every attribute (Section 3.3); the loadings profile is that of Figure 3.

its sales history lengthens (Bajari et al. 2019), in line with the argument that returns to data diminish by the law of large numbers (Varian 2019). By contrast, a model of health outcomes improves at an increasing rate, over a range, as researchers add attributes (Carballa-Smichowski et al. 2025), and a U.S. court found that rival search engines given access to Google's data would "keep getting better at a faster and faster rate up to some point" (United States v. Google LLC 2025b).

The debate treats data as a homogeneous input, whose quantity is the volume of a dataset—the number of entries it records. However, datasets of the same volume can have very different value depending on whether that volume comes from more observations or more attributes. This paper reconciles these findings with a theory of the value of data in which a dataset has three margins. A firm learns, from a sample of observations, the weights that map attributes into the variable it predicts, and applies these weights to the attributes of the individual it predicts for. Hence, there are three distinct inputs: (i) observations (*scale*); (ii) the attributes the firm records for them (*training scope*); and (iii) the attributes it observes for the individual it predicts for (*target scope*). Measuring data by volume conflates these three margins. Figure 1 illustrates the importance of distinguishing them: a dataset that records one attribute out of ten for many individuals is worth less than a fifth of a dataset of the same volume that records every attribute for a tenth as many individuals.

The paper characterizes the returns to each margin and how the margins interact. Its central result concerns learning. When attributes are many, a single scalar, learning precision, determines how well the firm learns each weight, and a learning multiplier links the weights: with each attribute it adds to training, the firm learns the weight of every attribute it already trains on more precisely. The returns to each margin, and the gains from combining datasets, depend on this multiplier.

Model. I assume that a Bayesian firm predicts a target variable under quadratic loss. The target variable is a linear function of many attributes, or covariates, with unknown weights, plus idiosyncratic noise. Attributes are independent, so all interactions between them arise from learning. Attributes differ in signal strength, which the firm knows and which I capture through their *loadings*: attributes with higher loadings account for more of the predictable variation in the target variable. The firm takes attributes in a fixed order, and Remark 1 extends the results to different orders for training and for prediction. A dataset design sets the three margins, and its value is the fall in expected squared prediction error when the firm predicts optimally, with the posterior mean.

Learning and deployment. The firm uses every attribute that it both trains on and observes for the individual it predicts for; I call these attributes *deployed*. The value of a design is the sum, over the deployed attributes, of each attribute's loading times the fraction of its weight that the firm learns. An attribute that the firm observes for the individual but never trains on is worthless, because the firm cannot tell whether the target variable rises or falls with it. An attribute that the firm trains on but does not observe for the individual still has value, because with it the firm learns the weights it applies more precisely.

Learning precision and the learning multiplier. The model becomes tractable when the number of potential attributes is large and the sample grows in proportion. The firm then learns each weight as if it observed a signal of that weight alone, with precision equal to the attribute's loading times a common scalar, *learning precision*. Learning precision is the number of observations per potential attribute, per unit of *learning noise*, the noise that the firm faces when it learns each weight. The two training margins, scale and training scope, therefore affect what the firm learns only through learning precision. Learning noise is the validation error of the firm's model, its out-of-sample prediction error for a new individual whose trained attributes it observes. It solves a fixed-point equation: the noise that the firm faces when it learns each weight is the error that the rest of its model leaves, and that error depends on how well the firm learns each weight.

The slope of the fixed-point map at learning noise defines the *learning multiplier*. When the firm adds an attribute to training, learning noise falls directly, by the part of the attribute's signal that the firm learns, and indirectly: with lower noise, the firm learns every trained weight better, so learning noise falls further, and the successive falls sum to a Keynesian multiplier. When the firm also predicts with the new attribute, its marginal value has two parts: its own deployed value and a gain on every attribute already trained. This inframarginal gain is an externality of each attribute on the learning of all the others. The learning multiplier is observable: it equals one plus the percentage fall in validation error when the sample grows by one percent. The learning multiplier is a sufficient statistic in the sense of Chetty (2009): whatever the strength of the attributes, the multiplier determines the scale economies with

which the firm produces precision and, when the firm predicts with every attribute it trains on, the share of the marginal value of an attribute that comes from improving the weights already trained.

Returns to observations. Learning precision grows more than proportionally with the sample, because the firm spends at most one degree of freedom on each weight: once it has learned a weight, it spends further observations on the others. Returns to observations can therefore increase in samples that are small relative to the number of attributes and when training attributes are sufficiently stronger than some of the attributes used in prediction. They diminish in large samples. Firms often predict with fewer attributes than they train on. A platform predicts for new users from the few attributes they reveal on arrival, yet trains on established users with long histories. An advertiser observes a purchase only after it shows the ad, so it can train on purchases but cannot predict with them. Researchers in machine learning call attributes available only for training *privileged information* (Vapnik and Vashist 2009). Their weights are nuisance parameters for the prediction, as covariates that predict the outcome are for an experimenter who estimates a treatment effect (Lin 2013). A firm that trains on strong attributes and predicts with weak ones spends its first observations mostly on the nuisance weights of the strong attributes, and a growing share of further observations on the weights of the weak attributes it predicts with, so an observation can be worth more than the previous one. When the firm instead predicts with its strongest attributes, it spends a growing share of each further observation on weaker attributes and on averaging out noise, and returns to observations diminish.

Returns to attributes. Through the learning multiplier, returns to attributes that the firm trains on and predicts with can also increase. If attributes have equal strength, returns to attributes increase at every scale. If each new attribute is weaker than the last, they can increase at intermediate scale, and they diminish in small and in large samples, as well as at every scale when noise is high.

Interactions. Two margins are complements when more of one raises the marginal value of the other. Target scope complements both training margins, because the firm learns the weights from its training data and applies them to the attributes it deploys. At fixed target scope, scale and training scope are complements whenever the firm has learned at most half of each deployed weight, and substitutes at large scale: when noise falls, the firm learns more from each further observation about a weight it has learned poorly, and less about a weight it has learned well, because less of that weight remains to learn. When the firm also predicts with each attribute it adds to training, the two margins can remain complements in large samples.

Implications. Section 5 applies the results to data-sharing remedies, the combination of datasets and privacy rules. Data favor concentration if a dataset is worth more to a firm that already holds data than to a firm that holds none. Independent attributes of the same individuals that the firm trains on and predicts with are worth more combined than apart in every dataset. Pooling samples of the same attributes can be superadditive in small samples, provided that the firm trains on attributes stronger than some it predicts with. In each case, superadditivity comes from learning, and it weakens in large samples. If a rival predicts with its strongest attributes, its returns to observations diminish, so shared observations are worth most to it on the tasks for which it has the fewest observations. Noise added to the target variable in training data to protect privacy costs more to a firm that trains on and predicts with more attributes, given its sample, and little to a firm that has learned nearly every weight. A privacy rule that imposes the same noise on every firm therefore narrows a lead in attributes and can widen a lead in observations, to the benefit of the firm with the most data.

Contribution and related literature. The paper contributes to three literatures.

First, it contributes to the economics of data. Most authors in this literature specify the value of data in reduced form and study the consequences for competition (Prüfer and Schottmüller 2021; Hagiu and Wright 2023; Guijarro et al. 2024; de Cornière and Taylor 2025) and growth (Jones and Tonetti 2020). Others model data as signals about a single unknown state (Farboodi et al. 2019; Farboodi and Veldkamp 2026; Veldkamp and Chung 2024). In both strands data have one dimension, and any increasing returns to data come from feedback between data and demand. I derive the value of data from the primitives of prediction and characterize its curvature along three margins. To my knowledge, this is the first theory in economics in which increasing returns to data, on both training margins, arise from learning alone, with independent attributes and an optimal predictor. In independent and contemporaneous work, Schäfer (2025) obtains increasing returns to attributes from their correlation, in a model in which the weights are known. Privacy regulation based on consent can favor large firms (Campbell et al. 2015; Johnson et al. 2023); here, a privacy rule that adds noise to every firm's training data can favor the firm with the most data, through learning alone.

Second, it contributes to the economics of information. Authors who study information markets model data as general information structures and study how sellers price them and buyers use them (Bergemann et al. 2018; Bergemann and Bonatti 2024); others study the externalities that arise when individuals' data are correlated (Choi et al. 2019; Acemoğlu et al. 2022; Bergemann et al. 2022). I fix the decision problem, prediction under quadratic loss. Radner and Stiglitz (1984) show that the marginal value of information is zero at no information, so the value of information is not concave; here the value of observations can be convex because the firm learns many weights from the same observations and predicts with only some of them. Börgers et al. (2013) characterize when two signals are complements in every decision problem; here attributes are complements because the firm learns a common mapping from

the same observations.

Third, it connects to high-dimensional statistics and to scaling laws in machine learning. The fixed-point equation that defines learning noise is the Silverstein–Bai equation for the spectrum of a sample covariance matrix (Silverstein and Bai 1995; Bai and Silverstein 2010), on which the high-dimensional asymptotics of ridge regression rest (Dobriban and Wager 2018; Hastie et al. 2022). I prove the representation directly, attribute by attribute; the proof allows the firm to deploy fewer attributes than it trains on and makes the learning multiplier explicit (Appendix D.1). Theorists of scaling laws use the same features in training and in prediction (Maloney et al. 2022; Bahri et al. 2024; Bordelon et al. 2024); returns to observations then diminish, as Proposition 4(i) shows for any profile. Once the attributes a firm trains on differ from those it predicts with, the returns to each margin depend on the composition of data, as researchers in machine learning have argued (Sorscher et al. 2022; Hooker 2025).

Roadmap. Section 2 presents the model, Section 3 derives the value of a design and the learning-noise representation, Section 4 characterizes the returns to each margin and their interactions, Section 5 draws implications for policy, and Section 6 concludes. The appendices contain the proofs, simulations at finite dimensions and with non-Gaussian and correlated data, additional results, and connections to statistics.

2 Model

A firm predicts a scalar target variable for one individual, the target individual. To do so, it observes the target variable and some covariates for a sample of other individuals, the training sample, and some covariates of the target individual. The firm chooses the size of the sample and how many covariates to observe for the sample and for the target individual. I ask how much the firm lowers its expected prediction error with each choice.

2.1 Data-Generating Process

There are K potential covariates. Individuals are indexed by $i = 0, 1, 2, \ldots$: the firm predicts for individual 0, the target individual, and trains on individuals $1, \ldots, N$.[2] For each individual i, the target variable is a linear combination

$$y_i = \boldsymbol{x}_i^\top \boldsymbol{\beta} + \eta_i,$$

where $\boldsymbol{x}_i \in \mathbb{R}^K$ is the covariate vector of individual i, $\boldsymbol{\beta} \in \mathbb{R}^K$ is a vector of unknown weights common to all individuals, and $\eta_i \in \mathbb{R}$ is an idiosyncratic error.

[2]Individuals are i.i.d. conditional on $\boldsymbol{\beta}$, so labeling the sample $1, \ldots, N$ is without loss of generality.

Weights. The weights are independent of the covariates and of the idiosyncratic errors, and they are mutually independent zero-mean Gaussians, $\boldsymbol{\beta} \sim \mathcal{N}\left(0, \sigma_\beta^2 \boldsymbol{I}_K\right)$. The variance $\sigma_\beta^2 > 0$ measures prior uncertainty about how covariates map into the target variable.

Covariates. Covariates are independent across individuals and across coordinates, with covariate-specific variances:

$$x_{ik} \sim \mathcal{N}\Big(0, \frac{\omega_k}{K}\Big), \qquad i = 0, 1, 2, \dots, \quad k = 1, \dots, K.$$

I call ω_k the *loading* of covariate k.[3] Covariates become available to the firm in a fixed order, indexed by k, and loadings along that order are generated by a *loading profile* $\omega : [0,1] \to [0, \bar{\omega}]$, a measurable function bounded by some $\bar{\omega} < \infty$ with $\int_0^1 \omega(u)\, du = 1$:

$$\omega_k \equiv K \int_{(k-1)/K}^{k/K} \omega(u)\, du, \qquad k = 1, \dots, K.$$

Thus $\omega(u)$ is the loading of the covariate at position u in the order, and the loadings satisfy $\omega_k \in [0, \bar{\omega}]$ and

$$\frac{1}{K} \sum_{k=1}^{K} \omega_k = 1 \qquad \text{for every } K.$$

The normalization is a choice of units.[4] The firm knows the loadings and the order. Both may be random, provided they are realized and observed before the firm chooses its design; I condition on them.

Covariates enter every design in this order, for training and for prediction alike. The profile can take any shape: loadings need not decline along the order. I interpret the order as the sequence in which data sources become available to the firm: the firm builds or acquires sources one at a time, apps join its ecosystem over time, and its engineers add features to a model in an order that often follows collection cost or engineering effort. Remark 1 shows that a design in which covariates become available for training and for prediction in different orders has the same value as a design with a single order, and Remark 3 shows that a firm free to choose the order would choose a nonincreasing one.

[3] Heterogeneity can also come from the prior on the weights. If $x_{ik} \sim \mathcal{N}(0, a_k/K)$ and $\beta_k \sim \mathcal{N}(0, b_k)$, rescaling each covariate by the known constant $\sqrt{a_k}$ leaves $\boldsymbol{x}_i^\top \boldsymbol{\beta}$ unchanged and yields a model with unit loadings and prior variances $a_k b_k$. The rescaling is known and invertible, so value is unchanged, and every result below holds with ω_k read as covariate k's signal contribution relative to the average covariate.

[4] Under this normalization, the total covariate variance $\sum_k \mathrm{Var}(x_{ik}) = \frac{1}{K} \sum_k \omega_k$ equals one for every K, so the covariates share a fixed total signal whatever their number, and ω_k measures the strength of covariate k relative to the average covariate. It implies $\bar{\omega} \geq 1$.

Errors. The idiosyncratic errors are i.i.d. across individuals, independent of everything else, and Gaussian: $\eta_i \sim \mathcal{N}(0, \sigma_\eta^2)$, with $\sigma_\eta^2 > 0$.[5]

Under these assumptions,

$$\mathrm{Var}(y_i) = \sigma_\beta^2 \frac{1}{K} \sum_k \omega_k + \sigma_\eta^2 = \sigma_\beta^2(1 + \xi), \qquad \xi \equiv \frac{\sigma_\eta^2}{\sigma_\beta^2},$$

which is the same for all individuals and independent of K. The parameter ξ is the noise-to-signal ratio, or *noise ratio*. The two variances enter every result only through ξ and the scale σ_β^2, which Section 3.3 normalizes to one.

Signal share. Covariate k accounts for a share ω_k/K of the signal σ_β^2. The *signal share* of the first fraction $t \in [0, 1]$ of covariates is

$$\Omega(t) \equiv \int_0^t \omega(u)\, du,$$

so the first T covariates carry exactly $\Omega(T/K)$. It is concave if and only if ω is nonincreasing. In the *isotropic benchmark*, $\omega \equiv 1$: all covariates are ex ante identical, $\Omega(t) = t$, and value has a closed form (Appendix C.3).

The model is indexed by K, and I suppress the dependence of ω_k on K. The finite-sample results of Sections 3.1 and 3.2 hold at fixed K. Beyond them, K serves two purposes. It sets units: Section 3.3 measures the sample size and the numbers of covariates the firm uses relative to K. It also indexes a sequence of environments: Section 3.3 lets $K \to \infty$ with these ratios fixed.

Known loadings, unknown weights. The firm knows how much variation each covariate carries. Weights have zero mean, so the firm cannot tell whether the target variable rises or falls with a covariate.[6] With independent covariates, the firm cannot use one covariate as a proxy for another, so all interactions between covariates arise from learning. Simulations with correlated covariates reproduce the patterns of the returns in Section 4 (Appendix B.5).

2.2 Dataset Design

A design specifies how many covariates, taken in order, the firm uses for training and for prediction. The firm chooses a *dataset design* $D \equiv (N, T, T_0) \in \mathbb{N}_0^3$, with $T, T_0 \leq K$, consisting

[5] The high-dimensional analysis of Section 3.3 uses a misspecification floor, defined there, that is bounded away from zero; positive noise guarantees this for every design. Without noise, the floor vanishes when the firm trains on every covariate, and several bounds in the proofs divide by σ_η^2. The finite-sample results of Sections 3.1 and 3.2 hold with $\sigma_\eta^2 = 0$.

[6] With a known nonzero prior mean, value gains a term independent of the training data, and the noise that the firm faces when it learns the trained weights rises. I measure the value that data add to prior knowledge. Given these second moments, the Gaussian specification has maximum entropy (Jaynes 1957).

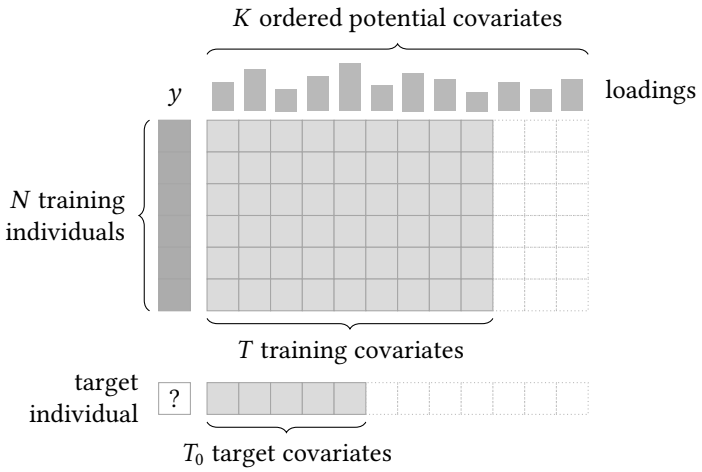


Figure 2. A dataset design $D = (N, T, T_0)$ with $K = 12$. Columns are the potential covariates in the order in which they become available; bars show their loadings. Shaded cells are observed: the target variable y and the first T covariates for each of the N training individuals, and the first T_0 covariates for the target individual, whose target variable the firm predicts.

of

(i) a *training sample size* N;

(ii) a number of *training covariates* T: the firm observes the target variable and the first T covariates for every individual in the sample;

(iii) a number of *target covariates* T_0: the firm observes the first T_0 covariates for the target individual.

Design D generates the dataset

$$\boldsymbol{d}_D \equiv \Big(\big(y_i, x_{i1}, \dots, x_{iT}\big)_{i=1}^{N},\ \big(x_{01}, \dots, x_{0T_0}\big) \Big),$$

which is empty for the empty design $D_\varnothing \equiv (0, 0, 0)$. Figure 2 depicts a design. When $T_0 < T$, the firm trains on covariates that it does not observe for the target individual, such as a purchase made after the firm predicts; following Vapnik and Vashist (2009), I call such covariates *privileged information*.

2.3 Prediction and the Value of a Design

Predictors and risk. A *predictor* for design D is a measurable function $\hat{y}$ of the dataset $\boldsymbol{d}_D$. The firm incurs quadratic loss $(y_0 - \hat{y})^2$. The *Bayes risk* of design D is the lowest expected loss any predictor for D achieves,

$$R(D) \equiv \inf_{\hat{y}} \mathbb{E}\big[(y_0 - \hat{y}(\boldsymbol{d}_D))^2\big].$$

Value. The *value* of design D is the fall in risk relative to predicting without data,

$$V(D) \equiv R(D_\varnothing) - R(D).$$

Order on designs. Designs are ordered componentwise: $D \leq D'$ if and only if $N \leq N'$, $T \leq T'$, and $T_0 \leq T_0'$. Because covariates enter every design in the same order, the firm observes under a larger design everything it observes under a smaller one.

3 The Value of a Design

I compute the value of a design in three steps. Under quadratic loss, the best predictor is the posterior mean (Section 3.1). Value is the sum, over the covariates the firm deploys, of each covariate's loading times the fraction of its weight that the firm learns (Section 3.2). When covariates are many, the firm faces the same noise, the validation error of its model, when it learns each weight, so one scalar determines how much it learns about each weight (Section 3.3). Proofs are in Appendix A.

3.1 The Optimal Predictor and Value

I first use only the quadratic loss.

Lemma 1 (Optimal predictor and value)**.** *For every design D:*

(i) the Bayes risk is attained by the posterior mean

$$\hat{y}_D^\star \equiv \mathbb{E}[y_0 \mid \boldsymbol{d}_D],$$

which is, almost surely, the unique optimal predictor, and the value of D is

$$V(D) = \mathrm{Var}(y_0) - \mathbb{E}\big[\,\mathrm{Var}(y_0 \mid \boldsymbol{d}_D)\big] = \mathrm{Var}(\hat{y}_D^\star);$$

(ii) value is nondecreasing: $V(D) \leq V(D')$ whenever $D \leq D'$.

Lemma 1 holds for any square-integrable target variable, whatever its relation to the covariates. Because the loss is quadratic, the optimal predictor is the posterior mean, and the value of a design is the expected fall in the variance of the target variable once the firm observes the dataset. Under a larger design, the firm observes everything it observes under a smaller one, so the larger design has at least as much value.[7]

[7] The optimal predictor therefore never overfits: its expected prediction error is nonincreasing in the sample size and in the number of covariates the firm observes.

3.2 Learning and Deployment

I now use the linear structure of the model. Write $\boldsymbol{y}_N \in \mathbb{R}^N$ and $\boldsymbol{X}_{N,T} \in \mathbb{R}^{N\times T}$ for the training sample $(y_i, x_{i1}, \dots, x_{iT})_{i=1}^N$, stacked by individual. The *posterior mean of the weights* is

$$\boldsymbol{m}_D \equiv \mathbb{E}\big[\boldsymbol{\beta} \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,T}\big] \in \mathbb{R}^K,$$

with $\boldsymbol{m}_D = \mathbf{0}$ when $N = 0$. With the training sample, the firm reduces its uncertainty about each weight. The *fraction learned* of covariate k's weight is the expected share of its prior variance that the firm removes with the training sample:

$$\mathcal{I}_k(D) \equiv 1 - \frac{\mathbb{E}\big[\operatorname{Var}(\beta_k \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,T})\big]}{\sigma_\beta^2} = \frac{\operatorname{Var}(m_{D,k})}{\sigma_\beta^2} \in [0,1].$$

Both objects depend on the design only through N and T.

Proposition 1 shows that, to predict, the firm multiplies each covariate it observes for the target individual by the posterior mean of that covariate's weight, and that the posterior mean of a weight the firm does not train on is its prior mean, zero. Only the first $T \wedge T_0 \equiv \min\{T, T_0\}$ covariates, which the firm both trains on and observes for the target individual, therefore enter the prediction. I call them the *deployed covariates*.

Proposition 1 (Learning and deployment)**.** *For every design $D = (N, T, T_0)$:*

(i) untrained weights have zero posterior means: $m_{D,k} = 0$ almost surely, and hence $\mathcal{I}_k(D) = 0$, for every $k > T$;

(ii) the optimal predictor applies the learned weights to the deployed covariates,

$$\hat{y}_D^\star = \sum_{k \le T\wedge T_0} x_{0k}\, m_{D,k};$$

(iii) value sums, over the deployed covariates, each covariate's loading times the fraction learned of its weight,

$$V(D) = \sigma_\beta^2 \frac{1}{K} \sum_{k \le T\wedge T_0} \omega_k\, \mathcal{I}_k(D),$$

and $\mathcal{I}_k(D)$ is nondecreasing in (N, T).

Proposition 1 holds whenever weights, covariates and errors are independent, centered and square-integrable; part (i), and with it the restriction to $k \le T \wedge T_0$ in parts (ii) and (iii), holds if weights and covariates are also symmetric about zero. Neither result uses Gaussianity.

Corollary 1 (Nested designs)**.** *For every design, $V(N, T, T_0) = V(N, T, T \wedge T_0)$: every design has the same value as a nested design, with $T_0 \le T$. I restrict attention to nested designs from now on.*

Proof. The two designs share the training sample, and therefore the fraction learned $\mathcal{I}_k$ of every covariate, and they have the same deployed covariates; they have the same value by Proposition 1(iii). □

In Proposition 1, learning and deployment enter separately. The training sample determines the fractions learned $\mathcal{I}_k$, how much the firm learns about how each covariate maps into the target variable. The target covariates determine which learned weights the firm applies, each in proportion to its loading ω_k. The firm can therefore add value in two ways. With a larger sample or an additional training covariate, it raises the fractions learned of the covariates it already deploys. It also deploys any covariate that it both trains on and observes for the target individual. A covariate available only for training has value only because, with it, the firm learns the weights of the deployed covariates more precisely. A covariate observed for the target individual but never trained on has none: the firm cannot tell whether the target variable rises or falls with it.

Remark 1 (Two orders). Covariates may become available for training in one order and for prediction in another, so that the firm trains on one set of covariates and observes another set for the target individual. The proof of Proposition 1 uses the covariates only through their independence, so value depends only on the sample size, on the covariates the firm trains on and on those it also observes for the target individual, the deployed covariates. If I list the deployed covariates first, the covariates available only for training next and the rest last, every such design becomes a nested design with the same value. I can therefore also interpret the single order as a ranking by use: covariates the firm observes both in its training sample and when it predicts, such as a user's location or device, come first; covariates available only for training, such as a purchase that the firm observes only after it shows an ad, come next. Appendix C.1 states the result and shows how Proposition 4 applies when the two orders differ.

What remains is to compute the fractions learned $\mathcal{I}_k(D)$.

3.3 The High-Dimensional Limit

At finite K, how much the firm learns about one weight depends on how much it learns about every other. When the number of covariates is large, the firm learns each weight as if that weight were the only unknown and the rest of the target variable were Gaussian noise; the variance of this noise, learning noise, is the same for every covariate.

Units. From now on I set $\sigma_\beta^2 = 1$, so that the predictable component of the target variable has unit variance, $\mathrm{Var}(\boldsymbol{x}_i^\top \boldsymbol{\beta}) = 1$.[8] Value is then the share of the predictable variance that the

[8]Dividing the target variable by σ_β divides every weight and the idiosyncratic error by σ_β and leaves ξ unchanged. The firm knows σ_β, so the rescaled data carry the same information, and value, a variance, is divided by σ_β^2.

firm explains with a design. Equivalently, it is the out-of-sample R^2 of the design divided by the attainable R^2, $R^2_{\max} = 1/(1+\xi)$, that of a firm that knows the weights and observes every covariate.

Design margins. I consider a sequence of nested designs $D_K = (N, T, T_0)$ indexed by K, and I suppress the dependence of N, T and T_0 on K.

Assumption 1 (Proportional regime)**.** *As $K \to \infty$,*

$$\frac{N}{K} \to n \in (0, \infty), \qquad \frac{T}{K} \to t \in [0, 1], \qquad \frac{T_0}{K} \to t_0 \in [0, t].$$

I refer to (n, t, t_0) as the three design margins: *training scale* n is the sample size relative to the number of potential covariates, *training scope* t is the fraction of potential covariates the firm trains on, and *target scope* t_0 is the fraction it deploys. From now on, scale refers to training scale n, and scope to training scope t. Training on the first fraction t of covariates leaves the *misspecification floor*

$$\underline{E}(t) \equiv \underbrace{\xi}_{\text{idiosyncratic noise}} + \underbrace{1 - \Omega(t)}_{\text{untrained covariates}},$$

the expected residual variance of the population regression of the target variable on the trained covariates. It is the firm's prediction error if it knew the weights of the trained covariates and deployed them all. Learning noise, defined below, never falls below the floor and approaches it as the sample grows.

Learning a single weight. Two functions describe how the firm learns in the limit. For a loading $\omega \geq 0$, a scale $n > 0$ and a noise level $E > 0$, let

$$\mathcal{I}(\omega; n, E) \equiv \frac{n\omega}{n\omega + E}, \qquad v(\omega; n, E) \equiv \omega\, \mathcal{I}(\omega; n, E) = \frac{n\omega^2}{n\omega + E}.$$

The first is the fraction of a weight that the firm learns when it observes the covariate and the target variable for $N = nK$ individuals and everything else in the target variable is Gaussian noise of variance E. Each observation carries precision $\omega/(KE)$ about the weight, so the sample carries $N\omega/(KE) = n\omega/E$, and the prior has precision one. At the same noise level, the firm learns the weights of high-loading covariates better. The second is the *deployed value* of the covariate: by deploying it, the firm explains a share of the signal equal to ω times the fraction of its weight that the firm has learned. Loading enters v twice: a stronger covariate carries more signal, and the firm learns its weight better.

Learning noise. In both functions, the noise level is a parameter. In the model, the noise that the firm faces when it learns one weight depends on the design: on which covariates

the firm trains on, and on how well it learns their weights. The noise has two parts: the misspecification floor $\underline{E}(t)$, and the *unlearned signal*, the signal of the other trained covariates that the firm has not learned, each covariate's loading times the unlearned fraction of its weight. Each covariate carries a vanishing share of the signal, so the noise is the same for every weight. If the firm learned every trained weight at noise E, the noise would be

$$\begin{aligned}\Psi(E; n, t) &\equiv \underbrace{\underline{E}(t)}_{\text{misspecification floor}} + \underbrace{\int_0^t \omega(u)\big(1 - \mathcal{I}(\omega(u); n, E)\big)\, du}_{\text{unlearned signal}} \\ &= \xi + 1 - \int_0^t v\big(\omega(u); n, E\big)\, du,\end{aligned} \tag{1}$$

where the second expression uses $\underline{E}(t) = \xi + 1 - \int_0^t \omega(u)\, du$. Thus $\Psi(E; n, t)$ is the prediction error of a model whose trained weights the firm learns at noise E, for a new individual whose trained covariates the firm observes. Since the noise that the firm faces for each weight is the error that the other learned weights leave, this noise should be a fixed point of Ψ. The next theorem proves that it is.[9] I extend $\Psi(\cdot; n, t)$ continuously to $E = 0$ by $\Psi(0; n, t) = \underline{E}(t)$ and write $\Psi'(E; n, t) \equiv \partial\Psi(E; n, t)/\partial E$.

Theorem 1 (Learning-noise representation)**.** *Under Assumption 1:*

(i) the equation

$$E = \Psi(E; n, t)$$

has a unique solution $E^\star(n, t)$, the learning noise*, which lies in $[\underline{E}(t), 1 + \xi]$ and satisfies $0 \le \Psi'(E^\star(n, t); n, t) < 1$;*

(ii) the fraction learned of every trained covariate converges to its single-weight value at learning noise, uniformly:

$$\max_{k \le T} \big|\mathcal{I}_k(D_K) - \mathcal{I}\big(\omega_k; n, E^\star(n, t)\big)\big| \longrightarrow 0;$$

(iii) the value of the design converges:

$$V(D_K) \longrightarrow \mathcal{V}(n, t, t_0) \equiv \int_0^{t_0} v\big(\omega(u); n, E^\star(n, t)\big)\, du.$$

I treat the limit as an approximation to a large but finite environment: in the simulations of Appendix B.2, it is within 1.1 percent of simulated value for every K from 100 to 1600.

When the margins are fixed, I abbreviate $E^\star \equiv E^\star(n, t)$ and write

$$\mathcal{I}^\star(u) \equiv \mathcal{I}\big(\omega(u); n, E^\star\big), \qquad v^\star(u) \equiv v\big(\omega(u); n, E^\star\big)$$

[9] Equation (1) is the Silverstein–Bai equation of random matrix theory (Appendix D.1).

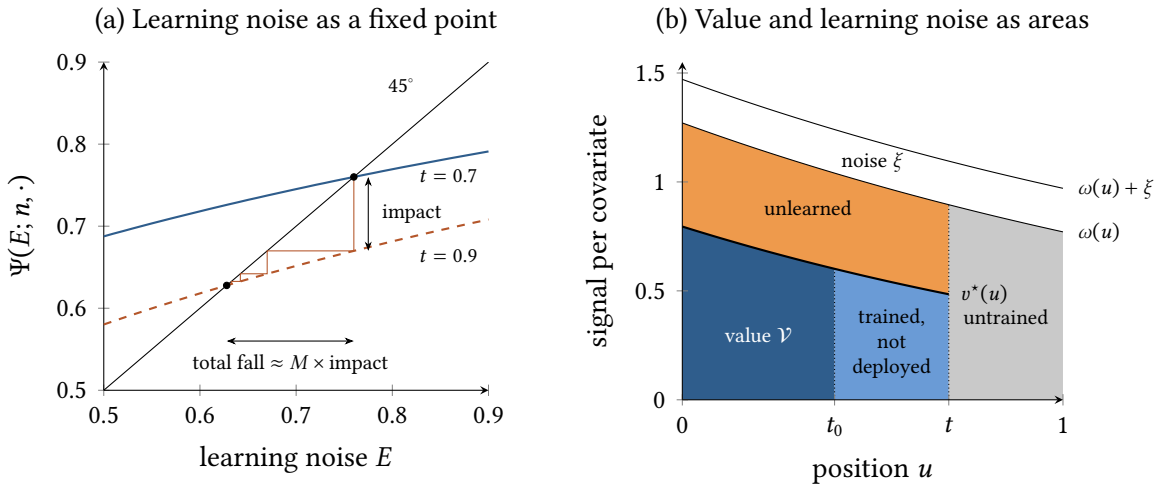


Figure 3. Learning noise. Exponential profile $\omega(u) = 0.5\, e^{-0.5u}/(1 - e^{-0.5})$, $\xi = 0.2$, $n = 1$; Theorem 1 holds for every profile. (a) Learning noise is the fixed point of $\Psi(\cdot; n, t)$, at $t = 0.7$ (solid) and $t = 0.9$ (dashed). Section 4.2 explains the staircase. (b) At $t = 0.7$ and $t_0 = 0.4$, deployed value $v^\star(u)$ lies below the loading $\omega(u)$, and the area under $\omega(u) + \xi$ is the variance of the target variable, $1 + \xi$. Value, $\mathcal{V} = 0.28$, is the area under $v^\star$ up to t_0. Learning noise, $E^\star = 0.76$, the fixed point in panel (a), is the sum of the unlearned, untrained and noise areas, as in Equation (1); the untrained and noise areas form the misspecification floor, $\underline{E}(t) = 0.45$. The trained, not deployed area is the value of the covariates that the firm trains on but does not observe for the target individual.

for the fraction learned and the deployed value of the covariate at position u. Figure 3(a) shows the map Ψ and its fixed point. Since $\mathcal{I}(\omega; n, E)$ depends on (n, E) only through n/E, the two training margins affect what the firm learns only through *learning precision*

$$\tau(n, t) \equiv \frac{n}{E^\star(n, t)} \ :$$

the firm learns the weight of a covariate with loading ω as if it observed one Gaussian signal of that weight alone, with precision $\tau\omega$. The prior has precision one, so the signal is as informative as $\tau\omega$ independent signals, each as precise as the prior, and the fraction learned, $\tau\omega/(1 + \tau\omega)$, is the share of the posterior precision that comes from the data.

Value is the area under the deployed-value curve $u \mapsto v^\star(u)$ up to target scope (Figure 3(b)). The curve depends on the training margins, on n directly and through $E^\star$, and on t through $E^\star$ alone; learning noise falls with both (Lemma 2). Target scope is the endpoint of the area. Since $v \leq \omega$, value never exceeds the signal share of the deployed covariates, $\mathcal{V}(n, t, t_0) \leq \Omega(t_0)$. Figure 3(b) also shows Equation (1) as areas: learning noise is the sum of the idiosyncratic noise, the signal of the untrained covariates and the unlearned signal of the trained ones.

The next corollary shows that the firm can observe learning noise.

Corollary 2 (Learning noise and prediction error)**.** *Under Assumption 1:*

(i) the prediction error of the fully deployed design converges to learning noise:

$$\lim_{K \to \infty} R(N, T, T) = E^\star(n, t), \qquad \mathcal{V}(n, t, t) = 1 + \xi - E^\star(n, t);$$

(ii) the prediction error of the design converges to

$$\lim_{K\to\infty} R(D_K) = E^\star(n,t) + \int_{t_0}^{t} v\big(\omega(u); n, E^\star(n,t)\big)\, du.$$

Proof. With $t_0 = t$, the integral in Theorem 1(iii) equals the integral in Equation (1) at $E = E^\star(n,t)$, so $\mathcal{V}(n,t,t) = 1 + \xi - \Psi(E^\star; n, t) = 1 + \xi - E^\star(n,t)$. Since $R(D) = \mathrm{Var}(y_0) - V(D)$ and $\mathrm{Var}(y_0) = 1 + \xi$, the limit of $R(N,T,T)$ is $E^\star(n,t)$, and the limit of $R(D_K)$ is $E^\star(n,t) + \mathcal{V}(n,t,t) - \mathcal{V}(n,t,t_0)$, which is part (ii). □

Learning noise is thus the *validation error* of the trained model: its out-of-sample prediction error for a new individual, drawn like the training sample, whose trained covariates the firm observes. The firm can measure this error on a held-out sample,[10] and it depends only on the training sample. The firm's prediction error for the target individual exceeds the validation error by the deployed value of the trained covariates that the firm does not observe for that individual.

The corollary matters for two reasons. First, it gives Equation (1) an economic meaning: the noise that the firm faces when it learns each weight is the validation error of the model that the learned weights form, so a better model lowers this noise for every weight. This feedback is the learning multiplier. Second, the firm observes validation error, so it can measure the multiplier from a learning curve without estimating the loadings, and the multiplier is a sufficient statistic for the spillovers among covariates (Section 4.2).

Ridge regression. In the limit, the optimal predictor is ridge regression on the trained covariates, with penalty equal to the misspecification floor (Step 2 of the proof of Theorem 1; Appendix B). The untrained covariates act as noise, so a firm that trains on fewer covariates shrinks its coefficients more. Because the posterior mean minimizes prediction error, this penalty is also the best one, and a firm that tunes a ridge penalty well attains the value in Theorem 1. A firm that tunes the penalty by cross-validation attains it in high dimensions (Patil et al. 2021).

Remark 2 (An algorithm that attains learning noise). A firm that uses approximate message passing estimates every weight from the residual that its current estimates of the other weights leave, corrected for the degrees of freedom it has spent on those estimates (Donoho et al. 2009). With posterior-mean updates, the noise that the firm faces at iteration ℓ follows $E_{\ell+1} = \Psi(E_\ell; n, t)$ and falls to learning noise, so the firm's predictions attain the value of the optimal predictor (Appendix D.2). When training scope rises and the firm starts from its old learning noise, the iterations trace the staircase of Figure 3(a).

[10] The firm can also compute it from the training sample alone. In the limit, learning noise equals the mean squared residual of the trained model divided by $(1 - \tilde{t}/n)^2$, where $K\tilde{t}$, the effective number of parameters, is the trace of the smoother matrix (Section 4.4): the generalized cross-validation criterion of Golub et al. (1979). This follows from the fixed point of approximate message passing (Appendix D.2).

Remark 3 (Which order?). Every result holds for any order in which covariates become available. A firm free to choose the order would take covariates in decreasing order of loading: replacing ω by its nonincreasing rearrangement weakly raises the signal share $\Omega(t)$ and the value $\mathcal{V}(n, t, t_0)$ of every design (Proposition 8 in Appendix C.2).

4 Returns to Data

The sign of the returns to each margin depends on four forces. Two lower returns: if loadings decline, each new covariate is weaker than the last; and the deployed value of a covariate is bounded by its loading, however well the firm learns its weight. Two raise returns: with each covariate it adds to training, the firm lowers learning noise and learns every weight better; and it spends fewer degrees of freedom on a weight it has partly learned. Returns to target scope depend only on the loading profile (Section 4.1). The training margins affect value through the learning multiplier (Section 4.2). Returns to scope increase if and only if the spillover rate exceeds the decay rate of loadings (Section 4.3). Returns to scale increase only if the firm trains on a covariate stronger than one it deploys, and only below some scale (Section 4.4). Section 4.5 studies how the margins interact, and Section 4.6 summarizes. From now on, I fix $n > 0$ and $0 < t_0 \leq t < 1$ with $\Omega(t_0) > 0$, so that the deployed covariates carry signal, assume that the profile is continuous wherever I differentiate with respect to t or t_0, and write $\tau \equiv \tau(n, t)$. Table 2 in Appendix A collects the notation.

4.1 Returns to Target Scope

Learning noise depends only on the training margins, so when the firm deploys one more covariate, value rises by that covariate's deployed value and every other term is unchanged. The returns to target scope therefore depend only on the loadings of the deployed covariates. If ω is differentiable and positive at t, let

$$\delta(t) \equiv -\frac{\omega'(t)}{\omega(t)}$$

be the *decay rate* of loadings at position t.

Proposition 2 (Returns to target scope)**.** *For every $t_0 \in (0, t]$,*

$$\frac{\partial \mathcal{V}}{\partial t_0} = v^\star(t_0) \geq 0,$$

with strict inequality if $\omega(t_0) > 0$. Moreover:

(i) if ω is differentiable and positive at t_0, then

$$\frac{\partial^2 \mathcal{V}}{\partial t_0^2} = -\big(2 - \mathcal{I}^\star(t_0)\big)\, v^\star(t_0)\, \delta(t_0) \ :$$

returns to target scope diminish at t_0 if $\delta(t_0) > 0$, are constant if $\delta(t_0) = 0$, and increase if $\delta(t_0) < 0$;

(ii) $\mathcal{V}$ is concave in t_0 on $[0, t]$ if and only if ω is nonincreasing on $[0, t]$.

The loading enters the deployed value $v^\star(t_0) = \omega(t_0)\,\mathcal{I}\big(\omega(t_0); n, E^\star\big)$ twice: a stronger covariate carries more signal, and the firm learns its weight better. The elasticity of deployed value with respect to the loading is therefore

$$\frac{\partial \ln v^\star}{\partial \ln \omega} = \underbrace{1}_{\text{signal}} + \underbrace{1 - \mathcal{I}^\star}_{\text{learning}} = 2 - \mathcal{I}^\star,$$

the factor that multiplies δ in Proposition 2. The elasticity exceeds one because a weaker covariate has lower deployed value for two reasons: it carries less signal, and the firm learns its weight less well ($\mathcal{I}^\star < 1$). As the firm learns every weight, $\mathcal{I}^\star \to 1$, and only the first reason remains. For two covariates with loadings $\omega < \omega'$, the ratio of their deployed values therefore rises with learning precision, from $(\omega/\omega')^2$ toward ω/ω'. Thus a firm with more training data values weak covariates relatively more.

4.2 The Learning Multiplier

Learning noise is a fixed point. When the firm expands a training margin, learning noise falls directly; the firm then learns every weight better, and learning noise falls again. Two scalars describe these falls and their value. The *absorptive capacity* of the deployed covariates,[11]

$$\mathcal{A}(n, t, t_0) \equiv \frac{1}{n} \int_0^{t_0} \mathcal{I}^\star(u)^2 \, du,$$

measures the value of a unit fall in learning noise at a given scale: by Equation (10), $\mathcal{I}^\star(u)^2/n$ is the increase in the variance that the covariate at position u explains when learning noise falls by one unit.

The absorptive capacity of the trained covariates is the slope of the fixed-point map, $\mathcal{A}(n, t, t) = \Psi'(E^\star; n, t)$, and it determines the *learning multiplier*

$$M(n, t) \equiv \frac{1}{1 - \Psi'(E^\star; n, t)} \in [1, \infty).$$

Suppose learning noise falls by one unit. With the trained covariates, the firm then explains Ψ' more of the variance of the target variable. Learning noise is the variance that the trained model leaves unexplained (Corollary 2), so it falls by a further Ψ'; after that fall, it falls by a further Ψ'^2, and the falls sum to $1 + \Psi' + \Psi'^2 + \cdots = M$. The sum is finite because $\Psi' < 1$ (Theorem 1(i)). The learning multiplier is a Keynesian multiplier, with Ψ' in place of the

[11] The term is from Cohen and Levinthal (1990).

marginal propensity to consume. Formally, let x be a training margin, n or t. Differentiating $E^\star = \Psi(E^\star; n, t)$ with respect to x gives

$$\frac{\partial E^\star}{\partial x} = \frac{\partial \Psi}{\partial x} + \Psi' \frac{\partial E^\star}{\partial x}, \qquad \text{so} \qquad \frac{\partial E^\star}{\partial x} = M \frac{\partial \Psi}{\partial x},$$

with Ψ and its derivatives evaluated at $E^\star$: the total effect of a training margin on learning noise is its partial effect at fixed noise, the *impact*, times the multiplier. The staircase of Figure 3(a) illustrates these falls. I suppress arguments when no confusion arises.

Lemma 2 (Marginal values)**.**

(i) $\dfrac{\partial E^\star}{\partial t} = -v^\star(t)\, M$ *and* $\dfrac{\partial E^\star}{\partial n} = -\dfrac{E^\star}{n}(M-1)$*;*

(ii) $\dfrac{\partial \mathcal{V}}{\partial t} = \mathcal{A}\, v^\star(t)\, M$*;*[12]

(iii) $\dfrac{\partial \mathcal{V}}{\partial n} = \mathcal{A}\, \dfrac{E^\star}{n} M.$

The two impacts in Lemma 2(i) have simple forms. When the firm trains on the covariate at position t, the misspecification floor falls by the covariate's loading $\omega(t)$, and the unlearned part of its weight, $\omega(t)\big(1 - \mathcal{I}^\star(t)\big)$, remains in the noise, so the impact of scope is $-v^\star(t)$. In Figure 3(b), a larger scale shrinks only the unlearned area, and a larger training scope also shrinks the untrained area; both raise the deployed value of every trained covariate. A one percent rise in scale lowers Ψ by Ψ' percent at fixed noise, so the elasticity of learning noise with respect to scale is $-\Psi' M = -(M-1)$.

A learning curve therefore measures the learning multiplier:

$$M = 1 + \alpha, \qquad \alpha \equiv -\frac{\partial \ln E^\star}{\partial \ln n},$$

where α is the percentage fall in validation error (Corollary 2) when the sample grows by one percent. The firm can estimate α by retraining its model on subsamples of its training data, with the same covariates, and by measuring the error on held-out individuals. The statistic α is sufficient in the sense of Chetty (2009): whatever the loading profile and the noise ratio, it determines the degree of scale economies in producing precision (Section 4.4) and, under full deployment, the share of spillovers in the marginal value of scope (Corollary 3) and the marginal value lost to noise in training targets (Corollary 5).

4.3 Returns to Training Scope

The firm can expand training scope with or without target scope. When the firm records a new covariate for all its users, it adds the covariate to the training sample and observes it for

[12] At $t_0 = t$, this is the right derivative: the firm adds a covariate to training without deploying it.

the target individual: under full deployment, $t_0 = t$ rises with t. When the covariate becomes available only after the firm predicts, as a purchase made after the firm shows an ad, the firm uses it only for training, and t rises at fixed $t_0 < t$ (Remark 1). I write $\bar{\mathcal{V}}(n,t) \equiv \mathcal{V}(n,t,t)$ for value under full deployment, call the curvature of $\bar{\mathcal{V}}$ in t the *returns to scope*, and call the curvature of $\mathcal{V}$ in t at fixed t_0 the *returns to training scope alone*. Under full deployment, the marginal value of scope has two parts.

Corollary 3 (Marginal and inframarginal value of scope)**.**

$$\frac{\partial \bar{\mathcal{V}}}{\partial t} = \underbrace{v^\star(t)}_{\textit{marginal}} + \underbrace{v^\star(t)\,(M-1)}_{\textit{inframarginal}} = v^\star(t)\,M.$$

The inframarginal share of the marginal value of scope is $\Psi'(E^\star; n, t) = \alpha/(1+\alpha)$.

Proof. The derivative is the sum of $\partial\mathcal{V}/\partial t_0$ (Proposition 2) and $\partial\mathcal{V}/\partial t$ (Lemma 2(ii)) at $t_0 = t$, where $\mathcal{A} = \Psi'(E^\star; n, t)$ and $\Psi' M = M - 1$. The share is $(M-1)/M = \Psi'$, and $M = 1 + \alpha$. □

The marginal effect is the new covariate's own deployed value at the current learning noise. The inframarginal effect is the *spillover*, the value the firm gains on the covariates it already trains: with the new covariate, learning noise falls, and the firm learns their weights better. For each unit of value that the firm gains directly from the marginal covariate, it gains $M - 1$ units on the inframarginal covariates. By Lemma 2(ii) and Proposition 2, a covariate that the firm adds to training is worth only its spillover, $\mathcal{A}\, v^\star M$, if the firm cannot observe it for the target individual, and $(1 + \mathcal{A}M)\, v^\star$ if it can.

The spillover share depends only on α, so a learning curve determines it without estimates of the loadings. Hestness et al. (2017) estimate α between 0.07 and 0.35 for deep networks on most translation, language, image and speech tasks. If the relation $M = 1 + \alpha$ held for these models, spillovers would account for 7 to 26 percent of the marginal value of scope.

Returns to scope increase if the marginal value of scope, $v^\star(t)M$, rises with scope. Two forces determine whether it does: spillovers and decay.

Spillovers. When the firm trains on the covariate at position t, it lowers learning noise by the proportion

$$\gamma(n,t) \equiv -\frac{\partial \ln E^\star}{\partial t} = \frac{\partial \ln \tau}{\partial t} = \frac{v^\star(t)\,M}{E^\star},$$

the covariate's *precision gain*: the proportional fall in validation error (Corollary 2) when the firm trains on the covariate, which equals its impact on learning noise, times the learning multiplier, relative to current noise. Lower noise raises the absorptive capacity of the trained covariates, $\Psi' = \frac{1}{n}\int_0^t \boldsymbol{\mathcal{I}}^\star(u)^2\,du$, which sums their contributions $\boldsymbol{\mathcal{I}}^\star(u)^2/n$: each is the spillover that a covariate receives when learning noise falls by one unit (Section 4.2). For $s \in (0,t]$ with

$\Omega(s) > 0$ and a function g on $[0, s]$, write

$$\langle g \rangle_s \equiv \frac{\int_0^s \mathcal{I}^\star(u)^2 \, g(u) \, du}{\int_0^s \mathcal{I}^\star(u)^2 \, du}$$

for the average of g over the first fraction s of covariates, with each covariate counted in proportion to its contribution to Ψ'. Thus $\langle \mathcal{I}^\star \rangle_s$ is the average fraction learned of these covariates, and $\langle 1 - \mathcal{I}^\star \rangle_s = 1 - \langle \mathcal{I}^\star \rangle_s$ the average fraction still to learn. The latter determines how fast absorptive capacity rises as noise falls (Lemma 3(d)):

$$\frac{\partial \ln \Psi'}{\partial \ln E} = -2 \langle 1 - \mathcal{I}^\star \rangle_t.$$

Lower noise raises the fraction learned of each weight with elasticity $1 - \mathcal{I}^\star$, the part still to learn, and the covariate's contribution, $\mathcal{I}^{\star 2}/n$, rises at twice that rate. Absorptive capacity therefore rises fastest when the firm has learned the trained weights poorly, and stops rising once the firm has learned them.

The *spillover rate* is

$$\Gamma(n, t) \equiv \frac{2\,\gamma(n, t)}{2 - \mathcal{I}^\star(t)} \Big[\underbrace{\big(1 - \mathcal{I}^\star(t)\big)}_{\text{marginal}} + \underbrace{(M - 1)\langle 1 - \mathcal{I}^\star \rangle_t}_{\text{inframarginal}} \Big] \; > \; 0 \quad \text{if } \omega(t) > 0.$$

Spillovers raise the marginal value of scope at the proportional rate $\big(2 - \mathcal{I}^\star(t)\big)\,\Gamma(n, t)$. The marginal term concerns the new covariate: lower noise raises its deployed value with elasticity $1 - \mathcal{I}^\star(t)$, and its contribution to absorptive capacity raises the multiplier; each effect equals $\gamma\big(1 - \mathcal{I}^\star(t)\big)$. The inframarginal term concerns the covariates already trained: lower noise raises their contributions, so absorptive capacity rises by the proportion $2\gamma\langle 1 - \mathcal{I}^\star \rangle_t$, and the multiplier by $M - 1$ times that proportion. The spillover rate depends on the profile only through its values on $[0, t]$.

Decay. By Section 4.1, deployed value has elasticity $2 - \mathcal{I}^\star(t)$ in the loading, so declining loadings lower the marginal value of scope at the proportional rate $\big(2 - \mathcal{I}^\star(t)\big)\,\delta(t)$. Both rates carry the factor $2 - \mathcal{I}^\star(t)$, so Proposition 3 compares Γ with δ directly.

Proposition 3 (Returns to scope)**.** *If ω is differentiable and positive at t:*

(i) the value of full deployment satisfies

$$\frac{\partial^2 \bar{\mathcal{V}}}{\partial t^2} = v^\star(t)\, M \big(2 - \mathcal{I}^\star(t)\big) \big[\Gamma(n, t) - \delta(t)\big],$$

so returns to scope increase at t if and only if $\delta(t) < \Gamma(n, t)$;

(ii) if $\delta(t) > 0$, there are $0 < \underline{n} \leq \bar{n}$ such that returns to scope diminish at t at every $n < \underline{n}$

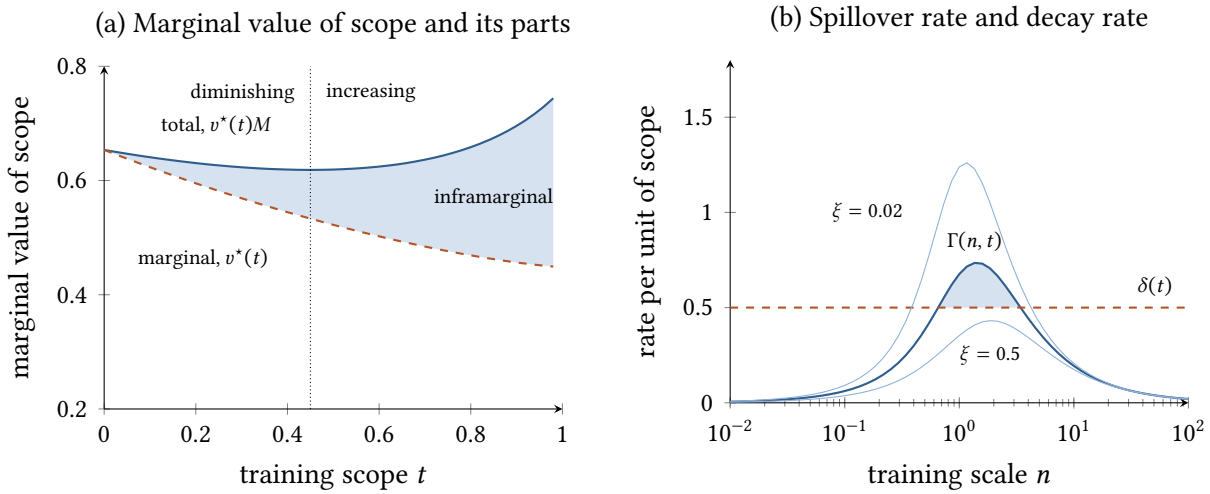


Figure 4. Returns to scope. Same example as Figure 3 ($\xi = 0.2$), full deployment. (a) The marginal value of scope at $n = 1$, its marginal part and its inframarginal part (shaded); returns to scope increase to the right of the dotted line. (b) The spillover rate $\Gamma(n,t)$ and the decay rate $\delta(t)$ at $t = 0.7$; shaded, $\Gamma > \delta$, increasing returns to scope. Thin curves, Γ at $\xi = 0.02$ and $\xi = 0.5$.

and every $n > \bar{n}$;

(iii) for covariates trained but not deployed ($t_0 < t$), $\partial^2\mathcal{V}/\partial t^2$ has the sign of

$$\Gamma(n,t) - \delta(t) + \frac{2\,\gamma(n,t)\,\langle 1 - \mathcal{I}^\star\rangle_{t_0}}{2 - \mathcal{I}^\star(t)};$$

(iv) if $\delta(t) > 0$, there is $\bar{\xi} > 0$ such that returns to scope diminish at t at every scale whenever $\xi > \bar{\xi}$.

Corollary 4 (Identical covariates)**.** *In the isotropic benchmark, $\bar{\mathcal{V}}$ is strictly convex in t at every scale.*

Part (i) compares the two rates at t: the spillover rate depends on the covariates trained up to t, and the decay rate on the slope of the profile at t, so the sign of the returns to scope can change as scope expands (Figure 4(a)). Wherever the profile is rising or flat, as within a cluster of covariates of similar strength, returns to scope increase at every scale. Corollary 4, the isotropic case, isolates spillovers.[13]

Part (ii) concerns scale. The firm gains from spillovers only if it has partly learned the trained weights. At small scale, the firm has learned little about any weight, so a new covariate lowers learning noise only slightly ($\gamma \to 0$). At large scale, a new covariate lowers learning noise substantially, but the firm has nearly learned the trained weights, and with

[13]Decay also vanishes if the firm adds covariates in random order. Each new covariate is then a random draw from the same distribution of loadings, and as $K \to \infty$, training on the first fraction t of covariates amounts to training on the fraction t of the covariates at every position, so value under full deployment is strictly convex in t at every scale (Appendix C.1). Carballa-Smichowski et al. (2025) add attributes of health data to a prediction model in random order and find returns to scope that first increase and then diminish. Spillovers can account for the increase; under a random order, the model predicts no decline.

lower noise it learns them only slightly better ($1 - \mathcal{I}^\star \to 0$ and $M \to 1$). The spillover rate therefore vanishes at small and at large scale, while decay remains at every scale: if loadings decline at t, returns to scope at t diminish at small and at large scale and can increase only at intermediate scale. Whether they increase at any scale depends on the profile and on noise. Noise weakens spillovers: a higher noise ratio raises learning noise and lowers the learning multiplier (Lemma 5), so each covariate lowers learning noise by a smaller proportion. In Figure 4(b), returns to scope increase at intermediate scale when $\xi = 0.02$ or $\xi = 0.2$ and diminish at every scale when $\xi = 0.5$, and by part (iv), they diminish at every scale whenever noise is high enough.

Part (iii) concerns privileged information: covariates available only for training. Such a covariate is worth only its spillover. Lower noise also raises the absorptive capacity of the deployed covariates; this effect is the positive last term in part (iii). At given n and t, returns to training scope alone therefore increase whenever returns to scope increase, and they can increase even if returns to scope diminish.

4.4 Returns to Scale

This subsection shows that returns to scale can increase even if observations are i.i.d., provided that some covariate that the firm uses only for training is stronger than one it deploys; they then increase only below some scale. Otherwise, returns to scale diminish at every scale (Proposition 4). Training scope is fixed, and I write $\underline{E} \equiv \underline{E}(t)$.

Scale affects value only through learning precision $\tau = n/E^\star$, so I treat precision as an intermediate good, which the firm produces from observations and turns into value:

$$n \xrightarrow{\text{cost of precision}} \tau \xrightarrow{\text{benefit of precision}} \mathcal{V}.$$

I first characterize the cost of precision, the number of observations the firm needs to attain it, and its scale economies; then the benefit of precision, the value the firm obtains from it. Average cost exceeds marginal cost, and marginal benefit diminishes.

The cost of precision. Since $\omega(1 - \mathcal{I}^\star) = \mathcal{I}^\star/\tau$, Equation (1) in its first form reads $E^\star = \underline{E} + \tilde{t}/\tau$, where the *learned scope*

$$\tilde{t}(\tau) \equiv \int_0^t \mathcal{I}^\star(u)\, du = \int_0^t \frac{\tau\omega(u)}{1 + \tau\omega(u)}\, du \;\le\; t$$

is t times the average fraction learned of the trained covariates. At given t, it depends on n only through τ, and if every trained covariate carries signal, $\tilde{t} \to t$ as $\tau \to \infty$. Multiplying this form of Equation (1) by $\tau = n/E^\star$ gives the number of observations the firm needs to attain

precision τ, its *cost of precision*:

$$n = C(\tau) \equiv \underbrace{\underline{E}\,\tau}_{\text{averaging out the floor}} + \underbrace{\tilde{t}(\tau)}_{\text{learning the weights}}. \tag{2}$$

Each observation adds one degree of freedom, and the firm allocates these degrees of freedom to two uses. Averaging out the floor costs $\underline{E}$ observations per unit of precision, a constant marginal cost. Learning the weights costs $\tilde{t}(\tau)$ observations per potential covariate: the degrees of freedom the firm spends on the weights, each counted by the fraction of it that the firm has learned. This cost is concave in τ, because the firm spends few further degrees of freedom on a weight it has mostly learned.

The cost of precision is therefore concave, and average cost exceeds marginal cost: the firm converts observations into precision with increasing returns. The degree of these returns, the ratio of average to marginal cost (Baumol et al. 1982), is the learning multiplier. Each fraction learned has elasticity $1 - \mathcal{I}^\star$ in τ, so $\tau\tilde{t}'(\tau) = \int_0^t \mathcal{I}^\star(1-\mathcal{I}^\star)\,du = \tilde{t} - n\Psi'$, and the marginal cost satisfies $\tau C'(\tau) = C(\tau) - n\Psi' = n(1-\Psi')$. Hence

$$\frac{C(\tau)}{\tau C'(\tau)} = \frac{\partial \ln \tau}{\partial \ln n} = M.$$

The learning multiplier thus also measures the scale economies in producing precision.

The benefit of precision. Value depends on the sample only through precision:

$$\mathcal{V} = \int_0^{t_0} \frac{\tau\,\omega(u)^2}{1+\tau\,\omega(u)}\,du.$$

The deployed value of each covariate is concave in τ and bounded by its loading: the firm gains less and less from further precision on a weight it has mostly learned. Returns to scale weigh the falling average cost of precision against this diminishing marginal benefit.

The marginal degree of freedom. To see how the two combine, consider how the firm allocates the degree of freedom that a marginal observation adds. Differentiating Equation (2) with respect to n gives

$$1 = \underbrace{\underline{E}\,\frac{\partial \tau}{\partial n}}_{\text{floor}} + \underbrace{\int_{t_0}^{t} q(u)\,du}_{\text{trained only}} + \underbrace{\int_0^{t_0} q(u)\,du}_{\text{deployed}}, \qquad q(u) \equiv \frac{\partial \mathcal{I}^\star(u)}{\partial n} = \frac{M\,\mathcal{I}^\star(u)\big(1-\mathcal{I}^\star(u)\big)}{n},$$

where $q(u)$ is the share of the marginal degree of freedom that the firm spends on learning weight u. The marginal value of an observation adds up the shares spent on the deployed

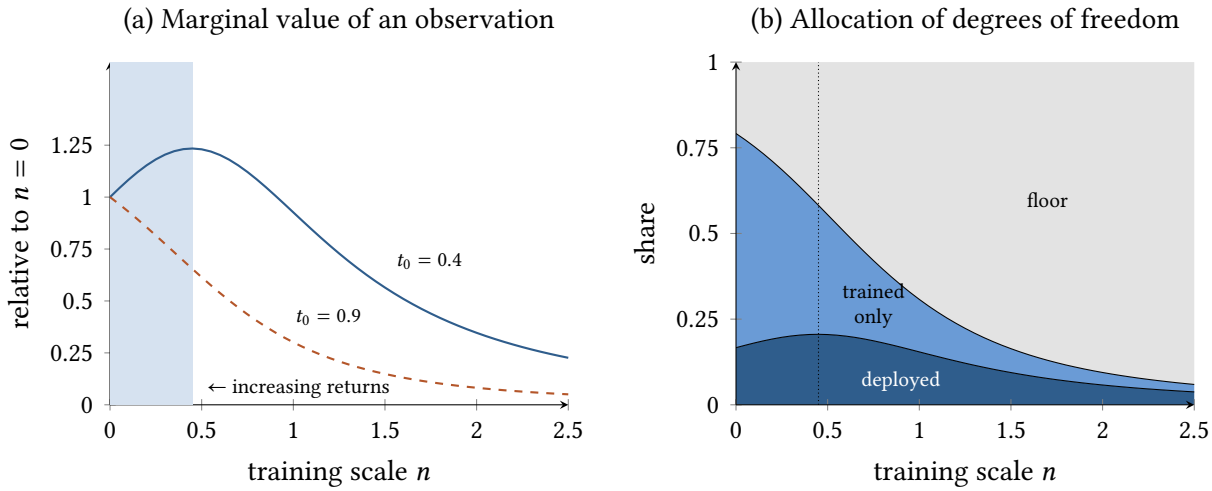


Figure 5. Returns to scale with privileged information. $\omega = 0.5$ on $[0, 0.4)$, 1.5 on $[0.4, 0.9)$ and 0.5 on $[0.9, 1]$; $\xi = 0.2$, $t = 0.9$. (a) Marginal value of an observation relative to its value at $n = 0$, when the firm deploys the weak covariates ($t_0 = 0.4$) and every covariate it trains ($t_0 = 0.9$); shaded, increasing returns to scale, up to $n = 0.45$ (dotted in panel b). (b) At $t_0 = 0.4$, the shares of the marginal degree of freedom spent on the floor, on the covariates trained only and on the deployed covariates.

weights, each multiplied by its loading:

$$\frac{\partial \mathcal{V}}{\partial n} = \int_0^{t_0} \omega(u)\, q(u)\, du.$$

Returns to scale therefore depend on how the shares move with the sample (proof of Proposition 4):

$$\frac{\partial \ln q(u)}{\partial \ln n} = 2M\Big[\underbrace{(M-1)\,\langle 1 - \mathcal{I}^\star\rangle_t}_{=\int_0^t q\,\mathcal{I}^\star\,du} - \mathcal{I}^\star(u)\Big]. \tag{3}$$

The first term is the average fraction learned across all uses of the marginal degree of freedom, each use counted by its share and the floor counted as unlearned. As the sample grows, the share of every weight learned beyond this average falls, and the share of every weight learned less rises. Since the firm learns strong weights first, it spends a decreasing fraction of its marginal degree of freedom on strong weights and an increasing fraction on weaker ones and, eventually, on the floor (Figure 5(b)).

Proposition 4 (Returns to scale)**.** *For every profile,*

$$\frac{\partial^2 \mathcal{V}}{\partial n^2} = \frac{2M}{n}\,\frac{\partial \mathcal{V}}{\partial n}\Big[(M-1)\,\langle 1 - \mathcal{I}^\star\rangle_t - \langle \mathcal{I}^\star\rangle_{t_0}\Big],$$

so returns to scale increase at n if and only if $(M-1)\langle 1 - \mathcal{I}^\star\rangle_t > \langle \mathcal{I}^\star\rangle_{t_0}$. Moreover:

(i) if the firm deploys its strongest trained covariates, $\omega(u) \geq \omega(u')$ whenever $u \leq t_0 < u' \leq t$, and in particular if it deploys every covariate it trains, returns to scale diminish at every $n > 0$;

(ii) if

$$\frac{\int_0^{t_0} \omega(u)^3 du}{\int_0^{t_0} \omega(u)^2 du} < \frac{\int_0^{t} \omega(u)^2 du}{1+\xi}, \tag{4}$$

there is $\hat{n} > 0$ such that returns to scale increase at every $n < \hat{n}$; if the reverse strict inequality holds, there is $\hat{n} > 0$ such that they diminish at every $n < \hat{n}$;

(iii) there is $n^\dagger > 0$ such that returns to scale diminish at every $n > n^\dagger$;

(iv) if $\underline{\omega} \equiv \inf_{u \le t} \omega(u) > 0$, returns to scale diminish at every $n > 0$ whenever $\xi > \sup_{u \le t} \omega(u)/\underline{\omega}$.

Deploying the strongest covariates. Varian (2019) attributes diminishing returns to the law of large numbers: each observation adds less precision than the last. Here each observation adds more precision than the last. If the firm deploys its strongest trained covariates, returns diminish (part i) because the allocation of the marginal degree of freedom shifts toward weaker covariates and the floor (Equation (3)). Estimated returns to data are concave in settings in which firms train and predict with the same covariates (Bajari et al. 2019; Schäfer and Sápi 2023; Lee and Wright 2025; Kaplan et al. 2020; Hoffmann et al. 2022), consistent with part (i).

Privileged information. If the firm trains on strong covariates and deploys weak ones, the shift of the marginal degree of freedom from strong to weak weights raises the marginal value of an observation. Part (ii) gives a sufficient condition for increasing returns to scale. At small scale, the fraction learned of each weight is proportional to its loading, $\mathcal{I}^\star(u) \approx n\omega(u)/(1+\xi)$, and by Equation (3) the share of the marginal degree of freedom spent on weight u rises with n if and only if $\omega(u) < \int_0^t \omega^2 du/(1+\xi)$. Inequality (4) states that an average of the deployed loadings lies below this threshold. The threshold falls with the noise ratio and rises with training scope. In the example of Figure 5, returns to scale increase up to $n = 0.14$ at $t = 0.7$, up to $n = 0.29$ at $t = 0.8$ and up to $n = 0.45$ at $t = 0.9$.

The condition of Proposition 4 also bounds how much the firm has learned: if returns to scale increase, then $\langle \mathcal{I}^\star \rangle_{t_0} < (M-1)\langle 1 - \mathcal{I}^\star \rangle_t < \alpha$. Increasing returns to scale therefore arise only early in learning, when the firm knows little about the weights it uses.

In large samples, the firm spends most of the marginal degree of freedom on the floor, and returns to scale diminish (part iii). Returns to scale also diminish at every scale if noise is high: once the noise ratio exceeds the ratio of the strongest to the weakest trained loading, the floor absorbs enough of each marginal degree of freedom (part iv). In between, the sign of the returns can change more than once, and returns can increase at intermediate scale even if inequality (4) fails.[14] Appendix B.4 shows that the increasing returns of Figure 5 appear

[14] Let $\xi = 0.02$, $t_0 = 0.3$ and $t = 0.9$, with $\omega = 3$ on $[0, 0.01)$, 0.2 on $[0.01, 0.3)$, 1.5 on $[0.3, 0.9)$ and 0.12 on $[0.9, 1]$. Returns to scale diminish up to $n = 0.39$, increase up to $n = 0.62$, and diminish beyond.

already with $K = 200$ covariates and hold with non-Gaussian data (Figure 8). Two covariates without noise show the mechanism in its simplest form (Appendix C.3).

4.5 Interactions between Margins

I call two margins complements when raising one raises the marginal value of the other. Proposition 5 shows that target scope complements both training margins. At fixed target scope, training scale and training scope are complements while the firm has learned little of each deployed weight, and substitutes in large samples. Under full deployment, the firm also deploys each covariate it adds to training, and scale and scope can remain complements in large samples.

Proposition 5 (Complements and substitutes)**.** *If ω is positive on $[0, 1]$:*

(i) target scope complements both training margins:

$$\frac{\partial^2 \mathcal{V}}{\partial n \, \partial t_0} = \frac{E^\star}{n} \frac{\mathcal{I}^\star(t_0)^2}{n} M > 0, \qquad \frac{\partial^2 \mathcal{V}}{\partial t \, \partial t_0} = \frac{\mathcal{I}^\star(t_0)^2}{n} v^\star(t) \, M > 0;$$

(ii) $\dfrac{\partial^2 \mathcal{V}}{\partial n \, \partial t} = \dfrac{\partial \mathcal{V}}{\partial n} \gamma(n, t) \, B(n, t, t_0)$*, where*

$$B(n, t, t_0) \equiv \underbrace{\langle 1 - 2\mathcal{I}^\star \rangle_{t_0}}_{\text{deployed weights}} + \underbrace{\big(1 - \mathcal{I}^\star(t)\big)}_{\text{marginal}} + \underbrace{2(M-1)\langle 1 - \mathcal{I}^\star \rangle_t}_{\text{inframarginal}};$$

moreover, at fixed target scope, training scale and training scope are complements whenever every deployed weight is at most half learned, $\mathcal{I}^\star(u) \leq 1/2$ for $u \leq t_0$, or equivalently $\max_{u \leq t_0} \omega(u) \leq 1/\tau(n, t)$, and they are substitutes at all sufficiently large n.

Target scope complements both training margins (part i) because learning and deployment enter separately in Proposition 1: the firm learns the weights from its training sample and applies them to the covariates it deploys, and the value of applying a weight rises with how well the firm has learned it.

Training scale and training scope interact through learning noise (part ii). When the firm trains on a new covariate, learning noise falls, and the sign of the interaction depends on whether lower noise raises the marginal value of an observation. A marginal observation raises the fraction learned of weight u by (Section 4.4)

$$q(u) = \frac{M}{n} \mathcal{I}^\star(u)\big(1 - \mathcal{I}^\star(u)\big),$$

which is proportional to the product of the fraction the firm has learned and the fraction it has yet to learn, largest at $\mathcal{I}^\star(u) = 1/2$. Lower noise raises $\mathcal{I}^\star(u)$, so it raises this product if $\mathcal{I}^\star(u) < 1/2$ and lowers it if $\mathcal{I}^\star(u) > 1/2$. If the firm has learned less than half of a deployed

weight, it learns more from each further observation when noise is lower; if it has learned more than half, it learns less, because less of the weight remains to learn. The first term of B averages this effect over the deployed weights. It is the only term when the firm learns one weight at a time, and it makes the two margins substitutes once the firm has learned most of each deployed weight.

Spillovers add a complementarity that learning one weight at a time misses. The other two terms of B come from the learning multiplier, and both are positive: by training on one more covariate, the firm raises the absorptive capacity of the trained covariates, Ψ', through which a larger sample lowers learning noise, and the lower noise raises Ψ' further. Each further observation is therefore worth more to a firm that trains on more covariates, and scale and training scope remain complements after the firm has learned half of every deployed weight.

At fixed target scope, training scale and training scope are thus complements at small and moderate scale and substitutes at large scale. Both margins raise every fraction learned, so the switch comes at a smaller scale when the firm trains on more covariates: in Figure 6(b), the boundary between complements and substitutes falls with training scope, and it rises with noise. Value depends on the training margins only through learning precision, so the contours of τ in the figure are isoquants: combinations of scale and training scope that yield the same value at fixed target scope. The half-learned condition in part (ii) fails first for the strongest deployed covariate, whose weight the firm learns fastest. The condition is sufficient: since the last two terms of B are positive, scale and training scope remain complements beyond the scale at which the firm has learned half of that weight. The model nests the complementarities between observations and covariates that Schäfer and Sápi (2023) and Lee and Wright (2025) document: at fixed target scope, they hold while the firm has learned the deployed weights poorly, and they give way to substitutability in large samples.

When the firm also deploys each covariate it adds to training, part (i) gives a second complementarity. For value under full deployment, $\bar{\mathcal{V}}$,

$$\frac{\partial^2 \bar{\mathcal{V}}}{\partial n \, \partial t} = \left[\frac{\partial^2 \mathcal{V}}{\partial n \, \partial t} + \frac{\partial^2 \mathcal{V}}{\partial n \, \partial t_0} \right]_{t_0 = t},$$

and the second term is positive. Scale and scope under full deployment can therefore remain complements in large samples. Among covariates of equal strength, they are complements in large samples if $t < (1+\xi)/2$ and substitutes if $t > (1+\xi)/2$ (Appendix C.3): a new covariate adds a weight to learn and lowers the floor. The noisier the target variable, the higher the floor and the wider the range of scopes at which the two margins are complements; if $\xi \geq 1$, they are complements at every scope.

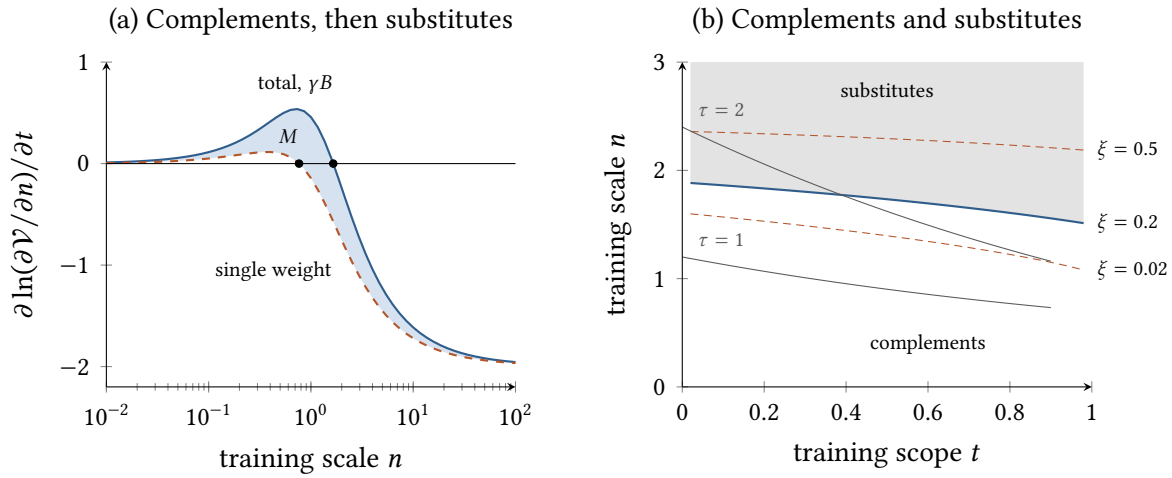


Figure 6. Interactions between scale and training scope at fixed target scope. Same example as Figure 3. (a) The effect of training scope on the marginal value of an observation at $t = 0.7$, with target scope fixed at $t_0 = t$: the total effect γB, its single-weight term $\gamma\langle 1 - 2\mathcal{I}^\star\rangle_t$, and, shaded, the two terms that come from the learning multiplier M; dots mark where the total and the single-weight term change sign. (b) The locus $B(n, t, t) = 0$, which separates complements (below) from substitutes (above; shaded at $\xi = 0.2$), at three noise ratios, and contours of learning precision τ at $\xi = 0.2$, along which value is constant at fixed target scope.

4.6 Summary

The returns to the training margins turn on three forces: spillovers, decay and the floor. Noise sets the strength of spillovers (Lemma 5) and the share of each observation that the firm spends on the floor. The shape of the loading profile sets decay and, when loadings rise along the order in which covariates become available, makes the covariates the firm uses only for training stronger than those it deploys. Table 1 crosses noise with the shape of the profile.

Table 1. Returns to each margin, by noise and loading profile

		Loadings		
Noise	Returns to	Flat	Declining	Rising
Any	target scope	constant	diminishing	increasing
Low	scope	increasing	increasing at intermediate n	increasing
	scale	diminishing	diminishing	increasing at small n
High	scope	increasing	diminishing	increasing
	scale	diminishing	diminishing	diminishing

Notes: Scope: full deployment. Scale: every $t_0 \leq t$. Sources: Propositions 2, 3 and 4, and Corollary 4. Declining, low noise: returns to scope diminish at small and at large n; in Figure 4(b), they increase at intermediate n when $\xi \leq 0.2$. Rising, low noise: with privileged information, if inequality (4) holds. High noise: $\xi > \bar{\xi}$ for scope, and ξ above the ratio of the strongest to the weakest trained loading for scale.

5 Implications

I now illustrate how the model can inform three policy questions: data-sharing remedies, the combination of datasets, and noise that a firm adds to its training data to protect privacy. For each, I describe the issue and the rule, map it into the model, and derive the implications.

Data-sharing remedies. *Shared observations help a rival most on the tasks for which its sample is smallest, and more so the more the rival observes about its users when it predicts.*

Article 6(11) of the Digital Markets Act requires gatekeeper search engines to share ranking, query, click and view data with rival search engines (European Parliament and Council of the European Union 2022), and the final judgment in the Google case requires Google to share search index and user-interaction data with qualified competitors (United States v. Google LLC 2025a). The question is where shared data help a rival most.

In the model, shared data raise the rival's scale, the number of observations per covariate for a given prediction task. A firm with many users can therefore have a small sample for a rare task: in *United States v. Google*, the court found that Google sees far more rare queries than its rivals, and that returns to user data diminish, but at a scale that remains uncertain (United States v. Google LLC 2024).

A rival that predicts with its strongest covariates gains most from shared observations on the tasks for which its sample is smallest, because its returns to observations diminish (Proposition 4(i)). Its predictions improve at an increasing rate with the data it receives only if it trains on covariates stronger than some it ranks results with, such as the queries a user makes later in the same session, and only below some number of observations per covariate (Proposition 4(i) and (iii)). Because target scope complements both training margins (Proposition 5(i)), shared training data are worth more to a rival that observes more about its own users when it predicts. For the same reason, when a rule limits what a firm may use at prediction time, as the EU rules on targeting studied by Goldfarb and Tucker (2011) do, the data the firm trains on are worth less, and so are its predictions.

Combining data. *Different covariates on the same individuals are worth more combined than apart in every dataset; samples of the same covariates are worth more combined only if the firm trains on covariates stronger than some it predicts with, and only in small samples.*

Firms combine datasets when they merge, buy data or share data under a remedy. For a given prediction task, the question is whether a dataset is worth more to a firm that already holds another than to a firm with no data, that is, whether combining is superadditive. Superadditivity is the sense in which data favor concentration.

In the model, the answer depends on which margin the second dataset expands. Datasets on the same individuals often record different covariates, such as a retailer's purchase records and a platform's browsing logs; combining them expands training and target scope. Datasets

on different individuals with the same covariates expand scale. Datasets of the first kind need not be initial segments of the order, so for this comparison only I describe a group of covariates by the set of their positions, $S \subseteq \{1, \dots, K\}$. Let $V_N(S)$ be the value when the firm observes the covariates in S for N training individuals and for the target individual.

Proposition 6 (Combining data)**.**

(i) For every K and N and all disjoint $S_1, S_2 \subseteq \{1, \dots, K\}$, $V_N(S_1 \cup S_2) \geq V_N(S_1) + V_N(S_2)$.

(ii) For all $n_1, n_2 > 0$, $\mathcal{V}(n_1+n_2, t, t_0) \geq \mathcal{V}(n_1, t, t_0)+\mathcal{V}(n_2, t, t_0)$ if $\mathcal{V}(n, t, t_0)/n$ is nondecreasing in n on $(0, n_1 + n_2]$, and the reverse inequality holds if $\mathcal{V}(n, t, t_0)/n$ is nonincreasing there.

By part (i), a firm that combines groups of covariates on the same individuals obtains economies of scope in the sense of Panzar and Willig (1981), in every finite dataset and whatever the strength of the covariates. The firm learns different weights from each group, and by training on each group it learns the weights of the other at least as well (Appendix A.3). Economies of scope in this sense hold even when returns to scope diminish (Proposition 3): returns to scope concern the next covariate in the order relative to the last, and economies of scope concern a combined dataset relative to its parts.[15]

Part (i) uses the independence of the covariates. With correlated groups, the firm can use each group as a partial proxy for the other: two perfectly correlated groups are worth no more combined than either alone. Correlation can thus raise the value of each group (Appendix B.5) and lower the gain from combining them.

From samples of individuals who carry the same covariates, the firm learns the same weights, so pooling them is superadditive if the average value of an observation rises with scale, and subadditive if it falls (part ii). When the firm predicts with its strongest trained covariates, value is concave in scale (Proposition 4(i)), so pooling samples is subadditive. When inequality (4) holds, value is convex at small scale (Proposition 4(ii)), so pooling two samples whose union is small enough is superadditive.

Data therefore favor concentration through covariates in every dataset, and through observations only if the firm trains on covariates stronger than some it predicts with, and then only in small samples. In both cases, superadditivity comes from learning, and it weakens in large samples: the gain from combining covariates vanishes as the firm learns every weight, and a large sample is worth less to a firm that holds another large sample of the same covariates than to a firm with no data.

Differential privacy. *A privacy rule that adds the same noise to every firm's training data narrows leads in covariates and can widen leads in observations, because a firm that has learned nearly every weight loses little.*

[15] When holders of different covariates on the same individuals combine their data, they must also divide the value. In the limit, the cooperative game among holders is convex, so its Shapley value lies in its core: every coalition of holders receives at least what it would obtain by combining its data on its own (Proposition 7 in Appendix C.1).

Privacy regulation may require a firm to add noise to the target variable in its training sample, as the Gaussian mechanism of differential privacy does (Dwork and Roth 2014).[16] The question is who bears the cost of the noise.

In the model, the firm observes the target variable of each training individual as $\tilde{y}_i = y_i + \zeta_i$, where the $\zeta_i \sim \mathcal{N}(0, \nu)$, $\nu > 0$, are i.i.d. and independent of everything else, while the target variable of the target individual is unchanged. Value depends on the training sample only through the fractions learned (Proposition 1), so the value of every design equals its value in the model with noise ratio $\xi + \nu$ (Appendix A.3). Let $E^\star_\nu$, M_ν and $\bar{\mathcal{V}}_\nu$ denote learning noise, the learning multiplier and value under full deployment in that model.

Corollary 5 (Differential privacy)**.** *If ω is positive on $[0, 1]$,*

$$\frac{\partial \bar{\mathcal{V}}_\nu}{\partial \nu} = -(M_\nu - 1).$$

The value lost to noise, $\bar{\mathcal{V}}_0 - \bar{\mathcal{V}}_\nu$, is positive, vanishes as $n \to 0$ and as $n \to \infty$, and strictly increases with training scope.

At $\nu = 0$, a marginal increase in noise lowers value by $M - 1 = \alpha$, the percentage fall in validation error when the sample grows by one percent (Section 4.2). The marginal cost of the noise to a firm is therefore minus the slope of its learning curve in logarithms. The loss is small when the firm has learned little and when it has learned nearly everything, so it is nonmonotone in scale. In the example of Figure 3, with $t = 0.7$ and $\nu = 0.1$, the loss rises with scale up to $n = 1.2$, about one observation per potential covariate, and falls beyond.

Suppose firms deploy every covariate they train and compete on the quality of their predictions, and a privacy rule imposes the same noise ν on the training data of every firm. The rule changes the gap in value between two firms by the difference between their losses. Of two firms with equal samples, the firm that trains on more covariates loses more value, so a lead in covariates narrows. Of two firms with different samples, the comparison depends on how much each has learned. Given the rival's sample, a firm whose sample is large enough loses less value than the rival, so a lead in observations widens, and the rule harms the smaller firm: in the example, a firm with $n = 10$ loses less than a quarter of what a rival with $n = 1.2$ loses. Given the leader's sample, a rival whose sample is small enough loses less value than the leader, and the lead narrows. A privacy rule can thus favor the firm with the most data, through learning alone.

[16] I thank Garrett Johnson for suggesting the connection with differential privacy. The formal guarantee of the mechanism requires a bounded target variable, so Corollary 5 measures the cost of the noise alone.

6 Conclusion

This paper develops a theory of the value of data in prediction. A dataset has three margins: the individuals a firm trains on, the attributes it records for them, and the attributes it observes when it predicts. When attributes are many, the training margins affect what the firm learns only through a single scalar, learning precision, which is inversely proportional to learning noise, the validation error of the firm's model. Learning noise falls as the firm learns each weight, and this feedback defines a learning multiplier: with each attribute it adds to training, the firm learns every weight it already trains on better.

Whether returns to data diminish depends on the margin of expansion, on noise and on the shape of the loading profile (Table 1). Returns to attributes that the firm trains on and predicts with weigh spillovers, through which each new attribute improves every trained weight, against the decline of loadings. They increase at every scale if loadings are flat or rise; if loadings decline, they can increase only at intermediate scale, and they diminish at every scale if noise is high. Returns to observations weigh the cost of precision, which rises less than proportionally because the firm spends at most one degree of freedom on each weight, against its benefit, which is concave because a deployed attribute is worth at most its loading. They diminish if the firm predicts with its strongest trained attributes, as with flat or declining loadings. They can increase below some scale, provided that the firm trains on attributes stronger than those it predicts with and noise is low enough.

Target scope complements both training margins. At fixed target scope, scale and training scope are complements whenever the firm has learned at most half of each deployed weight, and substitutes in large samples. When the firm also predicts with each attribute it adds to training, the two margins can remain complements in large samples; among attributes of equal strength, they do so at every scope if noise accounts for at least half of the variance of the target variable.

The learning multiplier equals one plus the percentage fall in validation error when the sample grows by one percent, so a learning curve measures it without knowledge of the loadings. It determines the scale economies with which the firm turns observations into precision and, when the firm predicts with every attribute it trains on, the share of the value of a new attribute that comes from the weights already trained and the marginal value the firm loses to privacy noise.

For competition policy, superadditivity, the sense in which data favor concentration, comes from learning. Independent attributes of the same individuals that the firm trains on and predicts with are worth more combined than apart in every dataset; pooling samples of the same attributes can be superadditive in small samples, provided that the firm trains on attributes stronger than some it predicts with. In each case, superadditivity weakens in large samples. Shared observations help a rival most on the tasks for which its sample is smallest, provided that it predicts with its strongest attributes.

The model also yields a cautionary result for privacy policy. A rule that adds the same noise to every firm's training data costs little to a firm that has learned little and to one that has learned nearly everything. Among firms with equal samples, the rule narrows a lead in attributes. Among firms with different samples, it widens the lead of a firm that has learned nearly every weight over rivals that have not, so a rule meant to protect users can favor the firm with the most data.

Three extensions follow. First, with prices for observations and attributes, the value function can serve as the benefit side of a firm's choice of dataset. Second, a dynamic model in which better predictions attract users, who supply further observations, and in which firms compete for these users, would show how the returns characterized here shape market structure. Third, the theory yields predictions that data can test: learning curves across prediction tasks would measure the learning multiplier, and prediction error can fall at an increasing rate only if the firm predicts with fewer attributes than it trains on, as for new users, and omits attributes stronger than some it keeps.

The central finding of this paper is that returns to data can increase through learning alone, with independent attributes, independent observations and an optimal predictor. With each attribute it adds to training, a firm learns every weight better, so an attribute can be worth more than the last; when the firm trains on attributes stronger than those it predicts with, an observation can be worth more than the previous one. A dataset can then be worth more to a firm that already holds data than to a firm that holds none, so the firms with the most data can gain the most from further data.

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A Proofs

Appendices A.1, A.2 and A.3 prove the results of Sections 3, 4 and 5, in order of appearance. Results stated only in later appendices are proved where they are stated. Each longer proof opens with its idea, and each step states what it shows. Table 2 collects the notation.

Table 2. Notation

Environment	
K	number of potential covariates
$\omega(u)$, ω_k	loading profile, with $\int_0^1 \omega = 1$ and $\omega \leq \bar{\omega}$, and loading of covariate k
$\Omega(t)$	signal share of the first fraction t of covariates, $\int_0^t \omega$
ξ	noise ratio, $\sigma_\eta^2/\sigma_\beta^2 = 1/R^2_{\max} - 1$
$R^2_{\max}$	highest attainable out-of-sample R^2, $1/(1+\xi)$
Designs and value	
$D = (N, T, T_0)$	sample size, number of training covariates, number of target covariates
(n, t, t_0)	training scale N/K, training scope T/K, target scope T_0/K
$R(D)$	Bayes risk of D, the lowest expected squared prediction error
$V(D)$	value of D: fall in Bayes risk relative to predicting without data
$\hat{y}^\star_D$, $\boldsymbol{m}_D$	optimal predictor, the posterior mean of y_0; posterior mean of the weights
$\boldsymbol{\mathcal{I}}_k(D)$	fraction learned of covariate k's weight: expected proportional reduction in its variance from the training sample
$\mathcal{V}(n, t, t_0)$	limiting value
Learning	
$\underline{E}(t)$	misspecification floor, $1 + \xi - \Omega(t)$
$\boldsymbol{\mathcal{I}}(\omega; n, E)$	fraction learned of a single weight, $n\omega/(n\omega + E)$
$v(\omega; n, E)$	deployed value of a covariate, $\omega\, \boldsymbol{\mathcal{I}}(\omega; n, E)$
$\Psi(E; n, t)$	fixed-point map, Equation (1)
$E^\star(n, t)$	learning noise, the fixed point of Ψ; validation error, the prediction error under full deployment
$\tau(n, t)$	learning precision, $n/E^\star$
$\boldsymbol{\mathcal{I}}^\star(u)$, $v^\star(u)$	fraction learned and deployed value of the covariate at position u, at $E^\star$
Returns	
$\delta(t)$	decay rate of loadings, $-\omega'(t)/\omega(t)$
$M(n, t)$	learning multiplier, $1/(1 - \Psi'(E^\star; n, t))$
$\mathcal{A}(n, t, t_0)$	absorptive capacity of the deployed covariates, $\frac{1}{n}\int_0^{t_0} \boldsymbol{\mathcal{I}}^{\star 2}$
α	percentage fall in validation error when scale grows by one percent, $-\partial \ln E^\star / \partial \ln n = M - 1$
$\bar{\mathcal{V}}(n, t)$	value under full deployment, $\mathcal{V}(n, t, t)$
$\gamma(n, t)$	precision gain from scope, $-\partial \ln E^\star / \partial t$
$\langle g \rangle_s$	average of g over $[0, s]$, weighted by $\boldsymbol{\mathcal{I}}^{\star 2}$
$\Gamma(n, t)$	spillover rate (Section 4.3)
$\tilde{t}(\tau)$	learned scope, $\int_0^t \boldsymbol{\mathcal{I}}^\star$
$C(\tau)$	cost of precision, the scale needed to attain precision τ, Equation (2)
$W(\tau)$	benefit of precision, value as a function of τ (proof of Proposition 4)
$q(u)$	share of the marginal degree of freedom spent on learning weight u, $\partial \boldsymbol{\mathcal{I}}^\star(u)/\partial n$
$\underline{\omega}$	weakest trained loading, $\inf_{u \leq t} \omega(u)$ (Proposition 4(iv))
$B(n, t, t_0)$	factor with the sign of the interaction between scale and training scope (Proposition 5)
Implications	
$V_N(\mathcal{S})$	value of training on and deploying the covariates in $\mathcal{S}$ with N training individuals (Proposition 6)
$\Pi(J)$	limit value of the combined covariates of the holders in coalition J (Proposition 7)
ν	variance of the noise added to the training targets (Corollary 5)

A.1 Proofs for Section 3

A predictor for design D is a function of the dataset $\boldsymbol{d}_D$, so conditioning on $\boldsymbol{d}_D$ means conditioning on everything the firm observes under D. Equalities between random variables hold almost surely.

Proof of Lemma 1. Under quadratic loss, the expected loss of any predictor splits into the expected loss of the posterior mean and the mean squared distance between the two predictions, so the posterior mean is the best predictor.

(i). The target variable has finite variance, $\mathrm{Var}(y_0) = \sigma_\beta^2(1+\xi)$. A predictor with infinite second moment has infinite expected loss, because Minkowski's inequality gives $\|\hat{y}\|_2 \leq \|y_0\|_2 + \|y_0 - \hat{y}\|_2$, so it suffices to consider predictors with a finite second moment. For any such predictor $\hat{y}$,

$$\mathbb{E}\big[(y_0 - \hat{y})^2\big] = \mathbb{E}\big[(y_0 - \hat{y}_D^\star)^2\big] + \mathbb{E}\big[(\hat{y}_D^\star - \hat{y})^2\big]. \tag{5}$$

The cross term vanishes: $\hat{y}_D^\star - \hat{y}$ is a function of the dataset, and the error of the posterior mean has mean zero given the dataset, so $\mathbb{E}\big[(y_0 - \hat{y}_D^\star)(\hat{y}_D^\star - \hat{y})\big] = \mathbb{E}\big[(\hat{y}_D^\star - \hat{y})\,\mathbb{E}[y_0 - \hat{y}_D^\star \mid \boldsymbol{d}_D]\big] = 0$. The second term in (5) is nonnegative and vanishes only if $\hat{y} = \hat{y}_D^\star$. Hence the posterior mean is the unique optimal predictor, and $R(D) = \mathbb{E}\big[(y_0 - \hat{y}_D^\star)^2\big] = \mathbb{E}\big[\mathrm{Var}(y_0 \mid \boldsymbol{d}_D)\big]$. Without data, the posterior mean is the prior mean $\mathbb{E}[y_0]$, and $R(D_\varnothing) = \mathrm{Var}(y_0)$, which gives the first expression for $V(D)$. Applying (5) to the constant predictor $\mathbb{E}[y_0]$, and using $\mathbb{E}[\hat{y}_D^\star] = \mathbb{E}[y_0]$, gives $\mathrm{Var}(y_0) = R(D) + \mathrm{Var}(\hat{y}_D^\star)$, so $V(D) = \mathrm{Var}(\hat{y}_D^\star)$.

(ii). If $D \leq D'$, the dataset $\boldsymbol{d}_D$ is a subvector of $\boldsymbol{d}_{D'}$, so every predictor for D is also a predictor for D'. The firm minimizes over a larger set under D', so $R(D') \leq R(D)$ and $V(D) \leq V(D')$. □

Proof of Proposition 1. The training sample cannot reveal the sign of a weight whose covariate it does not record, and the posterior mean of the target variable is the sum of the posterior means of its terms.

(i). Fix $k > T$ and flip the signs of β_k and of x_{ik} for every training individual $i \leq N$. The flip leaves each training target y_i unchanged, because it leaves $\beta_k x_{ik}$ and every other term unchanged, and it leaves the training covariates $\boldsymbol{X}_{N,T}$ unchanged, because the firm does not train on covariate k. Weights and covariates are independent and symmetric about zero, so the flip preserves their joint distribution. Hence $(\beta_k, \boldsymbol{y}_N, \boldsymbol{X}_{N,T})$ and $(-\beta_k, \boldsymbol{y}_N, \boldsymbol{X}_{N,T})$ have the same distribution, so $m_{D,k} = -m_{D,k} = 0$ and $\boldsymbol{\mathcal{I}}_k(D) = 0$.

(ii). Write $y_0 = \sum_k \beta_k x_{0k} + \eta_0$ and take expectations given the dataset term by term. The error η_0 has mean zero and is independent of the dataset, so it contributes zero. For $k > T_0$, the firm does not observe x_{0k}, which has mean zero and is independent of β_k and of the dataset, so the term contributes zero. For $k \leq T_0$, the firm observes x_{0k}, and the target covariates are independent of the weights and of the training sample, so $\mathbb{E}[\beta_k x_{0k} \mid \boldsymbol{d}_D] = x_{0k}\,\mathbb{E}[\beta_k \mid$

$\boldsymbol{y}_N, \boldsymbol{X}_{N,T}] = x_{0k} m_{D,k}$. Summing gives $\hat{y}_D^\star = \sum_{k \le T_0} x_{0k} m_{D,k}$, and by part (i), the terms with $k > T$ vanish.

(iii). By Lemma 1(i) and part (ii), $V(D) = \mathrm{Var}\left(\sum_{k \le T \wedge T_0} x_{0k} m_{D,k}\right)$. The target covariates are mutually independent, independent of the training sample, with mean zero and $\mathbb{E}[x_{0k}^2] = \omega_k / K$, so the cross terms vanish and

$$V(D) = \frac{1}{K} \sum_{k \le T \wedge T_0} \omega_k \, \mathbb{E}\left[m_{D,k}^2\right] = \sigma_\beta^2 \frac{1}{K} \sum_{k \le T \wedge T_0} \omega_k \, \mathcal{I}_k(D),$$

because $\mathbb{E}[m_{D,k}] = \mathbb{E}[\beta_k] = 0$ gives $\mathbb{E}[m_{D,k}^2] = \mathrm{Var}(m_{D,k}) = \sigma_\beta^2 \mathcal{I}_k(D)$. For monotonicity, let $N \le N'$ and $T \le T'$. The training sample under (N, T) is a subvector of that under (N', T'). The expected posterior variance of β_k is the lowest mean squared error of any estimate of β_k that is a function of the training sample, and the lowest error over a larger set of functions is weakly smaller, so $\mathcal{I}_k$ is nondecreasing in (N, T). □

Proof of Theorem 1. The proof makes the heuristic of Section 3.3 exact in five steps. Step 1 shows that the limit equation has a unique solution, at which the slope of Ψ is below one. Step 2 shows that the untrained covariates act as Gaussian noise whose variance is the misspecification floor, so that the posterior mean of the trained weights is ridge regression. Step 3 shows that the sample informs the firm about each trained weight as a single Gaussian signal of that weight would, with a noise level common to all weights. Step 4 counts degrees of freedom: the sample's N degrees of freedom are spent on averaging out the noise and on learning the weights, and this exact identity makes the common noise level solve the fixed-point equation up to an error that vanishes with K. Step 5 takes the limit.

Notation. Fix K and suppress it when no confusion arises. Write $\boldsymbol{X} \equiv \boldsymbol{X}_{N,T}$ for the training covariates, $\boldsymbol{A} \equiv \boldsymbol{X}^\top \boldsymbol{X}$, $\boldsymbol{x}_k$ for the k-th column of $\boldsymbol{X}$, $\boldsymbol{X}_{-k}$ for $\boldsymbol{X}$ without that column, and $\boldsymbol{z}_{\le T} \equiv (z_1, \ldots, z_T)$ for the first T coordinates of a vector $\boldsymbol{z}$. Let $n_K \equiv N/K$, and let $\bar{c} \equiv \underline{E}(T/K) = 1 + \xi - \Omega(T/K)$ be the misspecification floor at finite K, so that $\bar{c} \ge \xi > 0$ and $\bar{c} \to \underline{E}(t)$ by continuity of Ω. For K large, $n_K \in [n/2, 2n]$. The letter κ denotes a positive constant that depends only on σ_β^2, ξ, $\bar{\omega}$ and n; its value may change from line to line. Steps 2 to 4 keep σ_β^2 general; part (iii) is the case $\sigma_\beta^2 = 1$.

Step 1: the limit equation has a unique solution, and $\Psi' < 1$ there. Noisier learning leaves more signal unlearned, so Ψ rises with E, and it rises at a decreasing rate. For $E > 0$, $\Psi(\cdot; n, t)$ is continuously differentiable, with

$$\Psi'(E; n, t) = \int_0^t \frac{n\omega(u)^2}{(n\omega(u) + E)^2} \, du \ge 0, \qquad \frac{\partial^2 \Psi(E; n, t)}{\partial E^2} = -\int_0^t \frac{2n\omega(u)^2}{(n\omega(u) + E)^3} \, du \le 0.$$

Since $0 \le \Psi(E; n, t) - \underline{E}(t) = \int_0^t \omega E / (n\omega + E) \, du \le \int_0^t \min\{\omega, E/n\} \, du \to 0$ as $E \downarrow 0$, $\Psi(\cdot; n, t)$ is continuous on $[0, \infty)$, nondecreasing and concave, with $\Psi(0; n, t) = \underline{E}(t) \ge \xi > 0$ and $\Psi \le \underline{E}(t) + \Omega(t) = 1 + \xi$. The excess $\Phi(E) \equiv E - \Psi(E; n, t)$ is therefore convex, negative at

zero and unbounded above. A convex function that is negative at zero vanishes at most once, so Φ has a unique zero $E^\star(n,t)$, which lies in $[\underline{E}(t), 1+\xi]$ because Ψ maps into that interval. Convexity also gives $\Phi'(E^\star) \geq \big(\Phi(E^\star) - \Phi(0)\big)/E^\star = -\Phi(0)/E^\star > 0$, that is, $\Psi'(E^\star; n, t) < 1$: the map crosses the 45° line from above, so that

$$E < \Psi(E; n, t) \;\text{ for } E < E^\star(n,t), \qquad E > \Psi(E; n, t) \;\text{ for } E > E^\star(n,t). \tag{6}$$

The step uses only that ω is bounded and nonnegative.

Step 2: the untrained covariates act as Gaussian noise whose variance is the floor. Each training target is the trained part plus a residual that collects the untrained covariates and the idiosyncratic error,

$$y_i = \boldsymbol{x}_{i,\leq T}^\top \boldsymbol{\beta}_{\leq T} + w_i, \qquad w_i \equiv \sum_{k>T} \beta_k x_{ik} + \eta_i.$$

Given the untrained weights, the residuals are i.i.d. $\mathcal{N}(0, \sigma_\beta^2 c)$, independent of the trained weights and of $\boldsymbol{X}$, with

$$c \equiv \frac{1}{\sigma_\beta^2}\Big(\sigma_\eta^2 + \frac{1}{K}\sum_{k>T} \omega_k \beta_k^2\Big), \qquad \mathbb{E}[c] = \bar{c}, \qquad c \geq \xi.$$

A firm that knew c would face a Gaussian linear regression with known noise variance, and its posterior mean of the trained weights would be ridge regression with penalty c: for $\lambda > 0$, let $\tilde{\boldsymbol{m}}(\lambda) \equiv (\lambda \boldsymbol{I} + \boldsymbol{A})^{-1} \boldsymbol{X}^\top \boldsymbol{y}_N$ and $f_k(\lambda) \equiv \big[(\boldsymbol{I} + \boldsymbol{A}/\lambda)^{-1}\big]_{kk}$; then

$$\mathbb{E}[\boldsymbol{\beta}_{\leq T} \mid \boldsymbol{y}_N, \boldsymbol{X}, c] = \tilde{\boldsymbol{m}}(c), \qquad \mathrm{Var}(\beta_k \mid \boldsymbol{y}_N, \boldsymbol{X}, c) = \sigma_\beta^2 f_k(c).$$

The firm does not know c, but c averages $K - T$ independent terms, so it concentrates at $\bar{c}$; I now show that not knowing c costs the firm little. By the law of total variance, the expected posterior variance of β_k given the sample exceeds that given the sample and c by $\varepsilon_k \equiv \mathbb{E}\big[(\mathbb{E}[\beta_k \mid \boldsymbol{y}_N, \boldsymbol{X}, c] - m_{D,k})^2\big]$. The posterior mean $m_{D,k}$ is the best estimate of $\mathbb{E}[\beta_k \mid \boldsymbol{y}_N, \boldsymbol{X}, c]$ among functions of the sample, and $\tilde{m}_k(\bar{c})$ is one such function, so $\varepsilon_k \leq \mathbb{E}\big[(\tilde{m}_k(c) - \tilde{m}_k(\bar{c}))^2\big]$.

The ridge estimate moves little with the penalty. Its derivative is $\partial_\lambda \tilde{\boldsymbol{m}}(\lambda) = -\boldsymbol{p}(\lambda)$, with $\boldsymbol{p}(\lambda) \equiv (\lambda \boldsymbol{I} + \boldsymbol{A})^{-2} \boldsymbol{X}^\top \boldsymbol{y}_N$. Given $(\boldsymbol{X}, c)$, $\boldsymbol{y}_N \sim \mathcal{N}\big(0, \sigma_\beta^2 (\boldsymbol{X}\boldsymbol{X}^\top + c\boldsymbol{I}_N)\big)$, so $p_k(\lambda)$ is centered Gaussian with variance at most

$$\sigma_\beta^2 \max_{a \geq 0} \frac{a^2 + ca}{(\lambda + a)^4} \leq \sigma_\beta^2 \Big(\frac{1}{16\lambda^2} + \frac{27c}{256\lambda^3}\Big) \leq \kappa(1 + c) \qquad \text{for } \lambda \geq \xi.$$

By the Cauchy–Schwarz inequality along the segment between c and $\bar{c}$, both at least ξ, $(\tilde{m}_k(c) - \tilde{m}_k(\bar{c}))^2 \leq |c - \bar{c}| \left| \int_{\bar{c}}^{c} p_k(\lambda)^2 d\lambda \right|$. Taking expectations given $(\boldsymbol{X}, c)$ and then unconditionally,

$$\varepsilon_k \leq \kappa\, \mathbb{E}\big[(c - \bar{c})^2 (1 + c)\big] \leq \kappa \big(\mathbb{E}[(c - \bar{c})^4]\big)^{1/2} \big(\mathbb{E}[(1 + c)^2]\big)^{1/2} \leq \frac{\kappa}{K},$$

because $c - \bar{c} = \frac{1}{K}\sum_{k>T} \omega_k(\beta_k^2/\sigma_\beta^2 - 1)$ averages independent centered terms with bounded moments. The posterior variance also moves little with the penalty: $f_k'(\lambda) = \big[\boldsymbol{A}(\lambda \boldsymbol{I} + \boldsymbol{A})^{-2}\big]_{kk} \in [0, 1/(4\lambda)]$, so $|\mathbb{E} f_k(c) - \mathbb{E} f_k(\bar{c})| \le \mathbb{E}|c - \bar{c}|/(4\xi) \le \kappa/\sqrt{K}$. Since $\mathcal{I}_k(D) = 1 - \mathbb{E}[f_k(c)] - \varepsilon_k/\sigma_\beta^2$,

$$\max_{k \le T} \big|\mathcal{I}_k(D) - \mathcal{I}_k^\circ\big| \le \frac{\kappa}{\sqrt{K}}, \qquad \mathcal{I}_k^\circ \equiv 1 - \mathbb{E}\big[f_k(\bar{c})\big]. \tag{7}$$

The same bounds show that ridge regression with penalty $\bar{c}$ is within $\kappa K^{-1/2}$ of the posterior mean of each trained weight in root mean square. Up to an error of order $K^{-1/2}$, the firm thus learns the trained weights as if the untrained covariates were Gaussian noise whose variance is the misspecification floor.

Step 3: every trained weight faces the same learning noise. Let $\boldsymbol{R} \equiv (\bar{c}\boldsymbol{I}_N + \boldsymbol{X}\boldsymbol{X}^\top)^{-1}$ and $\boldsymbol{R}_{-k} \equiv (\bar{c}\boldsymbol{I}_N + \boldsymbol{X}_{-k}\boldsymbol{X}_{-k}^\top)^{-1}$, both with operator norm at most $1/\bar{c}$. By the Schur complement and the push-through identity,

$$f_k(\bar{c}) = \frac{1}{1 + q_k}, \qquad q_k \equiv \boldsymbol{x}_k^\top \boldsymbol{R}_{-k} \boldsymbol{x}_k, \qquad \text{so} \qquad \mathcal{I}_k^\circ = \mathbb{E}[h(q_k)], \quad h(q) \equiv \frac{q}{1+q}.$$

The scalar q_k is the precision of what the sample reveals about β_k once the firm accounts for the other trained weights. The prior has precision one, so the posterior precision is $1 + q_k$, and the fraction learned is the data's share of it, as in Section 3.3. The column $\boldsymbol{x}_k \sim \mathcal{N}(0, \frac{\omega_k}{K}\boldsymbol{I}_N)$ is independent of $\boldsymbol{R}_{-k}$, so q_k concentrates around its conditional mean: with $g_{-k} \equiv \frac{1}{N}\operatorname{tr} \boldsymbol{R}_{-k}$,

$$\mathbb{E}[q_k \mid \boldsymbol{X}_{-k}] = \omega_k \, n_K \, g_{-k}, \qquad \operatorname{Var}(q_k \mid \boldsymbol{X}_{-k}) = 2\Big(\frac{\omega_k}{K}\Big)^2 \operatorname{tr} \boldsymbol{R}_{-k}^2 \le \frac{2\bar{\omega}^2 N}{K^2 \bar{c}^2} \le \frac{\kappa}{K}.$$

The normalized trace $g \equiv \frac{1}{N}\operatorname{tr} \boldsymbol{R}$ is common to all weights, and it concentrates at its mean $\bar{g}_K \equiv \mathbb{E}[g]$. Removing one column changes $\operatorname{tr} \boldsymbol{R}$ by at most $1/\bar{c}$, so $|g - g_{-k}| \le 1/(N\bar{c})$. Replacing one column by an independent copy changes $\operatorname{tr} \boldsymbol{R}$ by at most $2/\bar{c}$, so the Efron–Stein inequality gives $\operatorname{Var}(g) \le 2T/(N^2\bar{c}^2) \le \kappa/K$. Since h is 1-Lipschitz on $[0, \infty)$ and $h(\omega_k n_K \bar{g}_K) = \mathcal{I}(\omega_k; n_K, 1/\bar{g}_K)$, the three bounds give, uniformly in $k \le T$,

$$\big|\mathcal{I}_k^\circ - \mathcal{I}(\omega_k; n_K, E_K)\big| \le \frac{\kappa}{\sqrt{K}}, \qquad E_K \equiv \frac{1}{\bar{g}_K}. \tag{8}$$

The firm thus learns every trained weight as if it observed a single Gaussian signal of that weight with noise E_K, the same for every weight. Since $\boldsymbol{R} \preceq \bar{c}^{-1}\boldsymbol{I}$, $E_K \ge \bar{c} \ge \xi$. Since $a \mapsto 1/(\bar{c} + a)$ is convex, Jensen's inequality, applied to the eigenvalues of $\boldsymbol{X}\boldsymbol{X}^\top$ and then to the expectation, gives $\bar{g}_K \ge 1/\big(\bar{c} + \mathbb{E}[\frac{1}{N}\operatorname{tr} \boldsymbol{X}\boldsymbol{X}^\top]\big) = 1/\big(\bar{c} + \Omega(T/K)\big)$, so $E_K \le 1 + \xi$.

Step 4: counting degrees of freedom pins down the learning noise. The firm spends the N degrees of freedom of its sample on averaging out the noise and on learning the weights. Formally, since $(\bar{c}\boldsymbol{I}_N + \boldsymbol{X}\boldsymbol{X}^\top)\boldsymbol{R} = \boldsymbol{I}_N$, taking traces gives $\bar{c}\operatorname{tr} \boldsymbol{R} + \sum_{j \le T} \boldsymbol{x}_j^\top \boldsymbol{R} \boldsymbol{x}_j = N$. By the Sherman–Morrison formula, $\boldsymbol{R} = \boldsymbol{R}_{-j} - \boldsymbol{R}_{-j}\boldsymbol{x}_j\boldsymbol{x}_j^\top \boldsymbol{R}_{-j}/(1 + q_j)$, so $\boldsymbol{x}_j^\top \boldsymbol{R} \boldsymbol{x}_j = h(q_j)$: the degrees

of freedom spent on weight j equal the fraction of it that the firm learns. Taking expectations and dividing by N gives the exact identity

$$\frac{\bar{c}}{E_K} = 1 - \frac{1}{N}\sum_{j\le T} \mathcal{I}_j^{\circ}. \tag{9}$$

Its limit is the cost of precision, Equation (2): multiplied by n_K, it reads $\bar{c}\, n_K/E_K + \frac{1}{K}\sum_{j\le T}\mathcal{I}_j^{\circ} = n_K$. Substituting (8), multiplying by $E_K \le 1+\xi$, and using $\frac{E}{m}\,\mathcal{I}(\omega;m,E) = \omega\big(1-\mathcal{I}(\omega;m,E)\big)$ gives

$$\big|E_K - \Psi_K(E_K)\big| \le \frac{\kappa}{\sqrt{K}}, \qquad \Psi_K(E) \equiv \bar{c} + \frac{1}{K}\sum_{j\le T}\omega_j\big(1-\mathcal{I}(\omega_j;n_K,E)\big).$$

The map Ψ_K is the fixed-point map of the finite model, the floor plus the unlearned signal of the trained covariates, so E_K solves the finite fixed-point equation up to a vanishing error.

Step 5: the finite fixed point converges to the limit one. It remains to show that Ψ_K converges to Ψ uniformly. Let $\omega_K(u) \equiv \omega_{\lceil uK\rceil}$ be the step function of cell averages, so that $\frac{1}{K}\sum_{j\le T} f(\omega_j) = \int_0^{T/K} f(\omega_K(u))\,du$ for every f. Averaging over the cells of a uniform partition is a contraction in $L^1[0,1]$ and converges uniformly for continuous functions, which are dense in L^1; hence $\|\omega_K-\omega\|_{L^1}\to 0$. The functions $f_{E,m}(\omega) \equiv \omega\big(1-\mathcal{I}(\omega;m,E)\big) = \omega E/(m\omega+E)$ satisfy $|\partial_\omega f_{E,m}|\le 1$ and are Lipschitz in (E,m) uniformly over $\omega\in[0,\bar{\omega}]$, $E\in[\xi,1+\xi]$ and $m\in[n/2,2n]$. Therefore

$$\sup_{E\in[\xi,1+\xi]}\left|\int_0^{T/K} f_{E,n_K}(\omega_K)\,du - \int_0^{t} f_{E,n}(\omega)\,du\right| \le \|\omega_K-\omega\|_{L^1} + \bar{\omega}\left|\frac{T}{K}-t\right| + \kappa|n_K-n| \longrightarrow 0,$$

which, with $\bar{c}\to \underline{E}(t)$, gives $\sup_{E\in[\xi,1+\xi]}|\Psi_K(E)-\Psi(E;n,t)|\to 0$ and hence $\Phi(E_K)\to 0$. Since $E_K\in[\xi,1+\xi]$ and Φ is continuous with the unique zero $E^\star(n,t)$, every limit point of (E_K) equals $E^\star(n,t)$, so $E_K\to E^\star(n,t)$.

Parts (ii) and (iii). Part (ii) follows from (7), (8), $E_K\to E^\star(n,t)$ and $n_K\to n$, because $(\omega,m,E)\mapsto\mathcal{I}(\omega;m,E)$ is uniformly Lipschitz on $[0,\bar{\omega}]\times[n/2,2n]\times[\xi,1+\xi]$. For part (iii), Proposition 1(iii) and nestedness give $V(D_K) = \sigma_\beta^2\frac{1}{K}\sum_{k\le T_0}\omega_k\mathcal{I}_k(D_K)$. By part (ii),

$$\left|V(D_K) - \sigma_\beta^2\,\frac{1}{K}\sum_{k\le T_0} v(\omega_k;n,E^\star)\right| \le \sigma_\beta^2\,\bar{\omega}\,\max_{k\le T}\big|\mathcal{I}_k(D_K)-\mathcal{I}(\omega_k;n,E^\star)\big| \longrightarrow 0.$$

Since $|\partial_\omega v(\omega;n,E)|\le 1$, the argument of Step 5 shows that the sum converges to $\int_0^{t_0} v(\omega(u);n,E^\star)\,du$. □

A.2 Proofs for Section 4

Throughout, $\sigma_\beta^2 = 1$; $n>0$ and $0<t_0\le t<1$ are fixed unless stated otherwise; and the profile is continuous wherever I differentiate in t or t_0. Then $\mathcal{V}(n,t,t_0) = \int_0^{t_0} v(\omega(u);n,E^\star(n,t))\,du$ is continuously differentiable in (n,t,t_0). The formula is also defined for $t_0>t$, so at $t_0=t$ the

right derivative in t at fixed t_0, the effect of training on one more covariate without deploying it, equals the derivative of this extension, and $\partial\bar{\mathcal{V}}/\partial t = [\partial\mathcal{V}/\partial t + \partial\mathcal{V}/\partial t_0]_{t_0=t}$ by the chain rule. An average $\langle\cdot\rangle_s$ appears only when $\Omega(s) > 0$, which makes it well defined, and $\Omega(t_0) > 0$ makes $\mathcal{A} > 0$. I write $\mathcal{I} = \mathcal{I}(\omega; n, E)$ and $v = \omega\mathcal{I}$ for the single-weight functions of Section 3.3. Their derivatives are elementary; the next lemma collects those used repeatedly.

Lemma 3 (Derivative identities)**.** *For $n > 0$ and $E > 0$, with $\omega \geq 0$ in (a), (c) and (e), $\omega > 0$ in (b), and a profile ω in (d):*

(a) $\dfrac{\partial\mathcal{I}}{\partial E} = -\dfrac{\mathcal{I}(1-\mathcal{I})}{E}$ *and* $\dfrac{\partial\mathcal{I}}{\partial n} = \dfrac{\mathcal{I}(1-\mathcal{I})}{n}$;

(b) $\dfrac{\partial \ln v}{\partial \ln \omega} = 2 - \mathcal{I}$ *and* $\dfrac{\partial \ln v}{\partial E} = -\dfrac{1-\mathcal{I}}{E}$;

(c) $\dfrac{\mathcal{I}^2}{n} = \dfrac{\omega\,\mathcal{I}(1-\mathcal{I})}{E}$;

(d) for $s \in (0, 1]$, $\dfrac{\partial\Psi'(E; n, s)}{\partial E} = -\dfrac{2}{nE}\displaystyle\int_0^s \mathcal{I}(\omega(u); n, E)^2\big(1 - \mathcal{I}(\omega(u); n, E)\big)\,du$;

(e) the deployed value satisfies

$$\frac{\partial v}{\partial E} = -\frac{\mathcal{I}^2}{n}, \qquad \frac{\partial v}{\partial n} = \frac{E}{n}\cdot\frac{\mathcal{I}^2}{n}. \tag{10}$$

Proof. For (a), $\partial\mathcal{I}/\partial E = -n\omega/(n\omega + E)^2$, $\partial\mathcal{I}/\partial n = \omega E/(n\omega + E)^2$ and $\mathcal{I}(1-\mathcal{I}) = n\omega E/(n\omega + E)^2$. For (b), $\ln v = \ln n + 2\ln\omega - \ln(n\omega + E)$. For (c), both sides equal $n\omega^2/(n\omega + E)^2$. For (d), $\Psi'(E; n, s) = \frac{1}{n}\int_0^s \mathcal{I}^2\,du$; differentiate under the integral and use (a). For (e), $\partial v/\partial E = -n\omega^2/(n\omega + E)^2$ and $\partial v/\partial n = \omega^2 E/(n\omega + E)^2$. □

Proof of Proposition 2. Learning noise depends only on the training margins, so deploying one more covariate adds that covariate's deployed value and leaves every other term unchanged: $\partial\mathcal{V}/\partial t_0 = v^\star(t_0)$, which is positive if and only if $\omega(t_0) > 0$.

(i). Along the order, deployed value changes only through the loading, with elasticity $2 - \mathcal{I}^\star$ (Lemma 3(b)). Hence

$$\frac{\partial^2\mathcal{V}}{\partial t_0^2} = \frac{\partial \ln v}{\partial \ln \omega}\,v^\star(t_0)\,\frac{\omega'(t_0)}{\omega(t_0)} = -\big(2 - \mathcal{I}^\star(t_0)\big)\,v^\star(t_0)\,\delta(t_0),$$

and $\big(2 - \mathcal{I}^\star(t_0)\big)v^\star(t_0) > 0$, so the second derivative has the sign of $-\delta(t_0)$.

(ii). $\mathcal{V}$ is concave in t_0 on $[0, t]$ if and only if its derivative $u \mapsto v(\omega(u); n, E^\star)$ is nonincreasing on $[0, t]$. Deployed value strictly increases with the loading, so this holds if and only if ω is nonincreasing on $[0, t]$. □

Proof of Lemma 2. A training margin $x \in \{n, t\}$ lowers learning noise directly, by its impact $\partial\Psi/\partial x$ at fixed noise, and indirectly, because lower noise lowers Ψ again. Differentiating the fixed point sums the two effects.

(i). Since ω is continuous, $\Psi(E;n,t) = 1 + \xi - \int_0^t v(\omega(u);n,E)\,du$ is continuously differentiable in (E,n,t). The impacts are $\partial\Psi/\partial t = -v(\omega(t);n,E)$, because the new covariate leaves the floor and the unlearned part of its signal stays in the noise, and $\partial\Psi/\partial n = -(E/n)\,\Psi'(E;n,t)$, by (10). Because $1 - \Psi'(E^\star;n,t) > 0$ (Theorem 1(i)), the implicit function theorem applies to $E = \Psi(E;n,t)$ and gives $\partial E^\star/\partial x = M\,\partial\Psi/\partial x$. With $\Psi' M = M - 1$, this gives (i).

(ii) and (iii). Value responds to the training margins through the deployed values. Differentiating $\mathcal{V} = \int_0^{t_0} v(\omega(u);n,E^\star)\,du$ under the integral, which the bounded derivatives in (10) permit because $E^\star \geq \xi$, gives $\partial\mathcal{V}/\partial t = -\mathcal{A}\,\partial E^\star/\partial t$ and $\partial\mathcal{V}/\partial n = \mathcal{A}\,E^\star/n - \mathcal{A}\,\partial E^\star/\partial n$, where $\mathcal{A}\,E^\star/n$ is the direct effect of scale at fixed noise. Substituting (i) proves both claims. □

How scope moves absorptive capacity and the multiplier. Two consequences of Lemma 2 recur below. When the firm trains on one more covariate, learning noise falls by the proportion γ. Lower noise raises the contribution $\mathcal{I}^{\star 2}/n$ of each trained covariate, at the rate $2(1-\mathcal{I}^\star)$ per unit of proportional fall in noise (Lemma 3(a) and (d)). Since $\mathcal{A} = \Psi'(E^\star;n,t_0)$ with t_0 fixed and $n\mathcal{A} = \int_0^{t_0} \mathcal{I}^{\star 2}\,du$, the absorptive capacity of the deployed covariates rises at the rate

$$\frac{\partial \ln \mathcal{A}}{\partial t} = \frac{1}{\mathcal{A}}\,\frac{\partial \Psi'(E^\star;n,t_0)}{\partial E}\,\frac{\partial E^\star}{\partial t} = \frac{2v^\star(t)M}{nE^\star\mathcal{A}}\int_0^{t_0} \mathcal{I}^{\star 2}(1-\mathcal{I}^\star)\,du = 2\gamma\,\langle 1-\mathcal{I}^\star\rangle_{t_0}. \tag{11}$$

The multiplier rises for two reasons: the new covariate adds its own contribution to absorptive capacity, and lower noise raises the contributions of the covariates already trained. Since $M = 1/\big(1-\Psi'(E^\star;n,t)\big)$ and $\partial\Psi'(E;n,t)/\partial t = \mathcal{I}(\omega(t);n,E)^2/n$,

$$\frac{\partial \ln M}{\partial t} = M\Big[\frac{\mathcal{I}^\star(t)^2}{n} + \frac{\partial\Psi'(E^\star;n,t)}{\partial E}\,\frac{\partial E^\star}{\partial t}\Big] = \gamma\,\big(1-\mathcal{I}^\star(t)\big) + 2\gamma\,(M-1)\,\langle 1-\mathcal{I}^\star\rangle_t, \tag{12}$$

where the first term uses Lemma 3(c) and the second uses $M\Psi' = M - 1$.

Proof of Proposition 3. The marginal value of scope is $v^\star(t)M$ (Corollary 3), so returns to scope increase if and only if $v^\star(t)M$ rises with t. Its proportional growth adds the growth of the new covariate's deployed value, which decay lowers and lower noise raises, to the growth of the multiplier.

(i). Scope moves both arguments of $v^\star(t) = v\big(\omega(t);n,E^\star(n,t)\big)$. By Lemma 3(b) and Lemma 2(i),

$$\frac{d\ln v^\star(t)}{dt} = \big(2-\mathcal{I}^\star(t)\big)\frac{\omega'(t)}{\omega(t)} - \frac{1-\mathcal{I}^\star(t)}{E^\star}\,\frac{\partial E^\star}{\partial t} = -\big(2-\mathcal{I}^\star(t)\big)\delta(t) + \gamma\,\big(1-\mathcal{I}^\star(t)\big).$$

Adding the growth of the multiplier, (12),

$$\frac{d}{dt}\ln\frac{\partial\bar{\mathcal{V}}}{\partial t} = -\big(2-\mathcal{I}^\star(t)\big)\delta(t) + 2\gamma\Big[\big(1-\mathcal{I}^\star(t)\big) + (M-1)\langle 1-\mathcal{I}^\star\rangle_t\Big] = \big(2-\mathcal{I}^\star(t)\big)\big[\Gamma(n,t)-\delta(t)\big],$$

and multiplying by $\partial\bar{\mathcal{V}}/\partial t$ gives the claim. The objects $E^\star(n,t)$, M, γ, $\mathcal{I}^\star(t)$ and $\langle\cdot\rangle_t$ depend

on ω only through its values on $[0, t]$, and so does $\Gamma(n, t)$.

(ii). Spillovers vanish at both ends of the scale: at small scale the firm learns so little that a new covariate barely lowers learning noise, and at large scale it has learned so much that lower noise adds little. Formally, since $2 - \mathcal{I}^\star(t) \geq 1$ and $\langle 1 - \mathcal{I}^\star \rangle_t \leq 1$, $0 < \Gamma \leq 2\gamma\big[(1 - \mathcal{I}^\star(t)) + (M - 1)\big] \leq 2\gamma M$. Recall that $E^\star \in [\xi, 1 + \xi]$ and $\omega \leq \bar{\omega}$. As $n \to 0$, $v^\star(t) \leq n\bar{\omega}^2/\xi$ and $\Psi'(E^\star; n, t) \leq n\bar{\omega}^2/\xi^2$, so $M \to 1$, $\gamma \to 0$ and $\Gamma \to 0$. As $n \to \infty$, $1 - \mathcal{I}^\star(t) = E^\star/(n\omega(t) + E^\star) \leq (1 + \xi)/(n\omega(t)) \to 0$ because $\omega(t) > 0$, and $\Psi'(E^\star; n, t) \leq t/n$, so $M \to 1$, while $\gamma = v^\star(t)M/E^\star \leq \bar{\omega}M/\xi$ stays bounded; hence $\Gamma \to 0$. If $\delta(t) > 0$, then $\Gamma(n, t) < \delta(t)$ at all sufficiently small and all sufficiently large n, and part (i) gives $\partial^2\bar{\mathcal{V}}/\partial t^2 < 0$ there.

(iii). A covariate trained but not deployed raises value only through spillovers: $\partial\mathcal{V}/\partial t = \mathcal{A}v^\star(t)M > 0$ (Lemma 2(ii)). Its proportional growth adds the growth of the absorptive capacity of the deployed covariates, (11), to the expression in part (i), so $\partial^2\mathcal{V}/\partial t^2$ has the sign of $(2 - \mathcal{I}^\star(t))(\Gamma - \delta) + 2\gamma\langle 1 - \mathcal{I}^\star \rangle_{t_0}$.

(iv). Noise weakens spillovers at every scale: it bounds both the multiplier and the precision gain. Since $E^\star \geq \xi$ and $v^\star(t) \leq \omega(t)$, $\gamma \leq \omega(t)M/\xi$. Since $\mathcal{I}^\star(u) \leq n\omega(u)/(n\omega(u) + \xi)$ and $n\omega^2/(n\omega + \xi)^2 \leq \omega/(4\xi)$, $\Psi'(E^\star; n, t) \leq \Omega(t)/(4\xi) \leq 1/(4\xi)$, so $M \leq 2$ whenever $\xi \geq 1/2$. Hence, by the bound in part (ii), $\Gamma(n, t) \leq 2\gamma M \leq 8\,\omega(t)/\xi$ at every $n > 0$ when $\xi \geq 1/2$, and $\Gamma(n, t) < \delta(t)$ at every $n > 0$ when $\xi > \bar{\xi} \equiv \max\{1/2,\, 8\,\omega(t)/\delta(t)\}$. Part (i) gives the claim. □

Proof of Corollary 4. In the isotropic benchmark, loadings do not decay, $\delta \equiv 0$, while spillovers are positive, $\Gamma > 0$, because $\gamma > 0$ and $\mathcal{I}^\star(t) < 1$. Proposition 3(i) gives $\partial^2\bar{\mathcal{V}}/\partial t^2 > 0$. □

Proof of Proposition 4. Scale affects value only through learning precision τ. Write value as the benefit of precision, $W(\tau)$, and the scale as its cost, $C(\tau)$ (Equation (2)). The marginal value of an observation is then $W'(\tau)/C'(\tau)$, and its slope compares how fast the marginal benefit and the marginal cost of precision fall. Both are integrals over weights of the *unlearned signal* of each weight, $x(u) \equiv \omega(u)\big(1 - \mathcal{I}(u)\big) = \omega(u)/\big(1 + \tau\omega(u)\big)$, and as precision rises, each integrand falls at the proportional rate $2x(u)$. The marginal cost therefore falls faster than the marginal benefit, and returns to scale increase, if and only if the marginal degree of freedom goes, on average, to uses with more unlearned signal than the deployed weights that create value. Throughout, $\underline{E} \equiv \underline{E}(t) \geq \xi > 0$, $\mathcal{I} \equiv \tau\omega/(1 + \tau\omega)$, which equals $\mathcal{I}^\star$ at $\tau = n/E^\star$, and $x \equiv \omega(1 - \mathcal{I})$, so that $\tau x = \mathcal{I}$ and $\partial x/\partial\tau = -x^2$.

Step 1: precision rises strictly and smoothly with scale. By Equation (2), $n = C(\tau) = \underline{E}\tau + \tilde{t}(\tau)$, where C is continuously differentiable, with $C(0) = 0$, $C(\tau) \to \infty$ as $\tau \to \infty$, and $C'(\tau) = \underline{E} + \int_0^t \omega(1 - \mathcal{I})^2 du \geq \underline{E} > 0$. Hence τ is a continuously differentiable and strictly increasing function of n, with $\tau \to 0$ as $n \to 0$ and $\tau \to \infty$ as $n \to \infty$. Since $\tau C'(\tau) = n/M$ (Section 4.4), $\partial \ln \tau/\partial \ln n = M$.

Step 2: the marginal value of an observation and its allocation. Value is $W(\tau) \equiv \int_0^{t_0} \omega \mathcal{I}\, du$, with $W'(\tau) = \int_0^{t_0} x^2 du > 0$ because $\Omega(t_0) > 0$. Hence $\partial \mathcal{V}/\partial n = W'(\tau)/C'(\tau) > 0$. Since $\partial \mathcal{I}/\partial \tau = \omega(1-\mathcal{I})^2$, a marginal observation raises the fraction learned of weight u by $q(u) = \omega(1-\mathcal{I})^2/C'(\tau)$; the floor absorbs $\underline{E}/C'(\tau)$; these shares sum to one; and $\partial \mathcal{V}/\partial n = \int_0^{t_0} \omega\, q\, du$. Since $\tau C'(\tau) = n/M$ and $\tau\omega(1-\mathcal{I}) = \mathcal{I}$, $q(u) = M\mathcal{I}^\star(u)\big(1-\mathcal{I}^\star(u)\big)/n$.

Step 3: returns to scale compare two averages of unlearned signal. Let

$$\bar{x}_W(\tau) \equiv \frac{\int_0^{t_0} x^2 \cdot x\, du}{W'(\tau)}, \qquad \bar{x}_C(\tau) \equiv \frac{\underline{E} \cdot 0 + \int_0^{t} \omega(1-\mathcal{I})^2 \cdot x\, du}{C'(\tau)}.$$

The first is the average unlearned signal of the deployed weights, each weighted by its contribution to the marginal benefit. The second is the average unlearned signal across the uses of the marginal degree of freedom, each weighted by its share, with the floor at $x = 0$. Since $\partial x^2/\partial \tau = -2x \cdot x^2$ and $\partial[\omega(1-\mathcal{I})^2]/\partial\tau = -2x \cdot \omega(1-\mathcal{I})^2$, the marginal benefit and the marginal cost fall at the proportional rates

$$\frac{W''(\tau)}{W'(\tau)} = -2\,\bar{x}_W(\tau), \qquad \frac{C''(\tau)}{C'(\tau)} = -2\,\bar{x}_C(\tau).$$

Differentiating $\partial \mathcal{V}/\partial n = W'/C'$ in n, with $d\tau/dn = M\tau/n$ by Step 1, gives

$$\frac{\partial^2 \mathcal{V}}{\partial n^2} = \frac{\partial \mathcal{V}}{\partial n}\,\frac{M\tau}{n}\Big(\frac{W''}{W'} - \frac{C''}{C'}\Big) = \frac{2M}{n}\,\frac{\partial \mathcal{V}}{\partial n}\,\tau\big(\bar{x}_C - \bar{x}_W\big),$$

so returns to scale increase at n if and only if $\bar{x}_C > \bar{x}_W$. In the notation of Section 4, $\tau x = \mathcal{I}$ gives $\tau\bar{x}_W = \int_0^{t_0} \mathcal{I}^{\star 3} du \big/ \int_0^{t_0} \mathcal{I}^{\star 2} du = \langle \mathcal{I}^\star \rangle_{t_0}$. Multiplying the numerator and the denominator of $\tau\bar{x}_C$ by τ and using $\tau\omega(1-\mathcal{I}) = \mathcal{I}$,

$$\tau\,\bar{x}_C = \frac{\int_0^{t} \mathcal{I}^{\star 2}(1-\mathcal{I}^\star)\, du}{\underline{E}\tau + \int_0^{t} \mathcal{I}^\star(1-\mathcal{I}^\star)\, du} = \frac{n\Psi'\langle 1-\mathcal{I}^\star\rangle_t}{n/M} = (M-1)\,\langle 1-\mathcal{I}^\star\rangle_t,$$

because $\tau C'(\tau) = n/M$, $n\Psi' = \int_0^{t} \mathcal{I}^{\star 2} du$ and $M\Psi' = M-1$. Substituting gives the formula of the proposition. The same computation for the share $q(u) = \omega(1-\mathcal{I})^2/C'$ gives $\partial \ln q(u)/\partial \ln \tau = 2\tau\big(\bar{x}_C - x(u)\big) = 2\big[(M-1)\langle 1-\mathcal{I}^\star\rangle_t - \mathcal{I}^\star(u)\big]$, and multiplying by $\partial \ln \tau/\partial \ln n = M$ gives Equation (3).

(i). If the firm deploys its strongest trained covariates, the marginal degree of freedom goes in part to weaker weights and to the floor, which have less unlearned signal than the deployed weights. Formally, $\bar{x}_C$ averages over three groups. On the deployed weights, its weights are those of $\bar{x}_W$ divided by ω; since x increases with ω, Chebyshev's integral inequality makes the average of x over this group at most $\bar{x}_W$. On the weights trained but not deployed, every value of x is at most every value on the deployed weights, because x increases with ω and these covariates are weaker, and hence at most $\bar{x}_W$. The floor has positive weight $\underline{E}/C'$ and

$x = 0 < \bar{x}_W$. Hence $\bar{x}_C < \bar{x}_W$ at every $n > 0$.

(ii). At small scale, the firm has learned almost nothing, so the unlearned signal of each weight is its loading. As $n \to 0$, $\tau \to 0$ and $x \to \omega$ uniformly on $[0, t]$. Hence $\bar{x}_W \to \int_0^{t_0} \omega^3 du / \int_0^{t_0} \omega^2 du$ and $\bar{x}_C \to \int_0^t \omega^2 du / \big(\underline{E} + \Omega(t)\big) = \int_0^t \omega^2 du/(1+\xi)$. The first limit is below the second if and only if inequality (4) holds. Both averages are continuous in n, so if the limits differ, there is $\hat{n} > 0$ such that $\bar{x}_C - \bar{x}_W$ has the sign of their difference at every $n < \hat{n}$.

(iii). At large scale, the firm has nearly learned every trained weight, and the floor absorbs nearly all of each marginal degree of freedom. As $n \to \infty$, $\tau \to \infty$ and $\mathcal{I}^\star \to 1$ wherever $\omega > 0$, so $\langle \mathcal{I}^\star \rangle_{t_0} \to 1$ by dominated convergence, while $(M-1)\langle 1 - \mathcal{I}^\star \rangle_t = \tau \bar{x}_C \le \int_0^t \mathcal{I}^\star (1 - \mathcal{I}^\star) du/(\underline{E}\tau) \le t/(4\underline{E}\tau) \to 0$. Hence there is $n^\dagger > 0$ such that $\partial^2 \mathcal{V}/\partial n^2 < 0$ at every $n > n^\dagger$.

(iv). With high noise, the floor absorbs enough of each marginal degree of freedom at every scale. Let $\bar{\omega}_t \equiv \sup_{u \le t} \omega(u)$. Since x increases with ω, every deployed weight has $x \ge \underline{\omega}/(1 + \tau\underline{\omega})$, so $\bar{x}_W \ge \underline{\omega}/(1 + \tau\underline{\omega})$. Since $C'(\tau) \ge \underline{E} \ge \xi$ and $\omega(1 - \mathcal{I})^2 x = \omega^2/(1+\tau\omega)^3 \le \omega^2/(1+\tau\underline{\omega})$,

$$\bar{x}_C \le \frac{1}{\xi} \int_0^t \frac{\omega^2}{1 + \tau\underline{\omega}}\, du \le \frac{\bar{\omega}_t}{\xi} \frac{1}{1 + \tau\underline{\omega}} \le \frac{\bar{\omega}_t}{\xi\, \underline{\omega}} \bar{x}_W,$$

using $\int_0^t \omega^2 du \le \bar{\omega}_t\, \Omega(t) \le \bar{\omega}_t$. If $\xi > \bar{\omega}_t/\underline{\omega}$, then $\bar{x}_C < \bar{x}_W$ at every $n > 0$.

The share of a single weight at small scale (Section 4.4). As $n \to 0$, $M - 1 = M\Psi'$ and $\Psi' = \tau^2 \int_0^t \omega^2 du/n + o(\tau)$ with $n/\tau = E^\star \to 1 + \xi$, so $(M-1)/\tau \to \int_0^t \omega^2 du/(1+\xi)$, while $\mathcal{I}^\star(u)/\tau \to \omega(u)$ and $\langle 1 - \mathcal{I}^\star \rangle_t \to 1$. By Equation (3), the share of weight u therefore rises with n at all sufficiently small scales if $\omega(u) < \int_0^t \omega^2 du/(1+\xi)$ and falls if $\omega(u) > \int_0^t \omega^2 du/(1+\xi)$. □

Proof of Proposition 5. (i). Deploying the covariate at t_0 adds its deployed value, $v^\star(t_0)$ (Proposition 2). Each training margin raises this value by lowering learning noise, and scale also raises it directly. By (10) and Lemma 2(i),

$$\frac{\partial^2 \mathcal{V}}{\partial n\, \partial t_0} = \underbrace{\frac{E^\star}{n} \frac{\mathcal{I}^\star(t_0)^2}{n}}_{\text{direct}} + \underbrace{\frac{\mathcal{I}^\star(t_0)^2}{n} \frac{E^\star}{n} (M-1)}_{\text{through noise}} = \frac{E^\star}{n} \frac{\mathcal{I}^\star(t_0)^2}{n} M,$$

$$\frac{\partial^2 \mathcal{V}}{\partial t\, \partial t_0} = \frac{\partial v}{\partial E} \frac{\partial E^\star}{\partial t} = \frac{\mathcal{I}^\star(t_0)^2}{n} v^\star(t)\, M,$$

and both are positive because $\omega(t_0) > 0$.

(ii). By Lemma 2(iii), the marginal value of an observation is $\mathcal{A}\,(E^\star/n)\, M$: the absorptive capacity of the deployed covariates, times $E^\star/n$, the fall in noise that has the same effect on learning as one more observation, times the multiplier. Training scope moves all three factors. It raises absorptive capacity, (11); it lowers learning noise by the proportion γ; and it raises the multiplier, (12). Hence

$$\frac{\partial}{\partial t} \ln \frac{\partial \mathcal{V}}{\partial n} = 2\gamma \langle 1 - \mathcal{I}^\star \rangle_{t_0} - \gamma + \gamma\big(1 - \mathcal{I}^\star(t)\big) + 2\gamma(M-1)\langle 1 - \mathcal{I}^\star \rangle_t = \gamma\, B(n, t, t_0),$$

because $\langle\cdot\rangle_{t_0}$ is an average, so that $2\langle 1-\mathcal{I}^\star\rangle_{t_0}-1=\langle 1-2\mathcal{I}^\star\rangle_{t_0}$. Since $\partial\mathcal{V}/\partial n>0$ and $\gamma>0$, the cross-partial has the sign of B.

If every deployed weight is at most half learned, $\mathcal{I}^\star(u)\le 1/2$ on $[0,t_0]$, then $\langle 1-2\mathcal{I}^\star\rangle_{t_0}\ge 0$ and $B\ge 1-\mathcal{I}^\star(t)>0$; and $\mathcal{I}^\star(u)\le 1/2$ if and only if $\omega(u)\le E^\star/n=1/\tau$. In large samples, the firm has nearly learned every trained weight and the multiplier tends to one. Formally, ω is continuous and positive on $[0,t]$, so $\omega\ge\underline{\omega}$ there for some $\underline{\omega}>0$, and $1-\mathcal{I}^\star(u)\le(1+\xi)/(n\underline{\omega})\to 0$ uniformly on $[0,t]$; and $\Psi'(E^\star;n,t)\le t/n$, so $M\to 1$. Hence $B\to -1$, and the two margins are substitutes at all sufficiently large n. □

A.3 Proofs for Section 5

Proof of Proposition 6. (i). Value sums, over the deployed covariates, each loading times the fraction learned of its weight, and training on more covariates teaches the firm every weight at least as well. Formally, the proofs of Lemma 1 and Proposition 1 use the covariates only through their independence, never through their order. Proposition 1(iii) therefore holds for any set $\mathcal{S}\subseteq\{1,\dots,K\}$ of covariates observed for both the sample and the target individual: $V_N(\mathcal{S})=\frac{1}{K}\sum_{k\in\mathcal{S}}\omega_k\,\mathcal{I}_k(N,\mathcal{S})$, where $\mathcal{I}_k(N,\mathcal{S})$ is the fraction learned of covariate k's weight when the firm trains on $\mathcal{S}$ with N individuals. Training on $\mathcal{S}_1\cup\mathcal{S}_2$, the firm observes everything it observes when it trains on $\mathcal{S}_i$ alone, so the monotonicity argument in the proof of Proposition 1(iii) gives $\mathcal{I}_k(N,\mathcal{S}_1\cup\mathcal{S}_2)\ge\mathcal{I}_k(N,\mathcal{S}_i)$ for $k\in\mathcal{S}_i$, $i=1,2$. Splitting the sum over $\mathcal{S}_1\cup\mathcal{S}_2$ into its parts over $\mathcal{S}_1$ and over $\mathcal{S}_2$ gives the claim.

(ii). If the average value of an observation rises with scale, each sample is worth at most its share of the value of the pooled sample. Formally, let $f(n)\equiv\mathcal{V}(n,t,t_0)$ and $n_{12}\equiv n_1+n_2$. If $f(n)/n$ is nondecreasing on $(0,n_{12}]$, then $f(n_i)\le n_i f(n_{12})/n_{12}$ for $i=1,2$, and summing gives $f(n_1)+f(n_2)\le f(n_{12})$; if $f(n)/n$ is nonincreasing there, the inequalities reverse. Since $f(n)\to 0$ as $n\to 0$, convexity of f on $(0,n_{12}]$ makes $f(n)/n$ nondecreasing there, and concavity makes it nonincreasing, which gives the applications in Section 5. □

Proof of Corollary 5. Noise added to the training targets is indistinguishable, for the firm, from idiosyncratic noise, so the model with privacy noise is the model with noise ratio $\xi+\nu$. The derivative then follows from how learning noise responds to noise, and the loss vanishes at both ends of the scale because the learning multiplier tends to one there.

Value with noisy training data. The noisy training targets are $\tilde{y}_i=\boldsymbol{x}_i^\top\boldsymbol{\beta}+(\eta_i+\zeta_i)$, where the $\eta_i+\zeta_i$ are i.i.d. $\mathcal{N}(0,\xi+\nu)$ and independent of $\boldsymbol{\beta}$ and the covariates. The joint distribution of the weights and the training data is therefore that of the model with noise ratio $\xi+\nu$, and so is every fraction learned $\mathcal{I}_k(D)$. The target variable of the target individual affects value only through Proposition 1(ii)–(iii), whose proofs use only that η_0 has mean zero and is independent of the data. Hence $V(D)=\frac{1}{K}\sum_{k\le T\wedge T_0}\omega_k\mathcal{I}_k(D)$ is the value of D in the model with noise ratio $\xi+\nu$.

The derivative. Let $E^\star_\rho$, τ_ρ and M_ρ denote learning noise, learning precision and the learning multiplier at noise ratio $\xi + \rho$. By the fixed point, $\bar{\mathcal{V}}_\nu = \int_0^t v^\star = 1 + \xi + \nu - E^\star_\nu$, and Lemma 5 in Appendix C.2 gives $\partial E^\star_\nu / \partial \nu = M_\nu$. Hence $\partial \bar{\mathcal{V}}_\nu / \partial \nu = 1 - M_\nu$, and the loss is $\bar{\mathcal{V}}_0 - \bar{\mathcal{V}}_\nu = \int_0^\nu (M_\rho - 1)\, d\rho$.

The loss is positive and rises with training scope. At every noise ratio, $M = 1/(1 - \Psi')$ with $\Psi' = \frac{1}{n}\int_0^t \mathcal{I}^\star(u)^2\, du > 0$, since ω is positive; each M_ρ therefore exceeds one, and the loss is positive. Training scope raises each M_ρ, as in (12): the total derivative of Ψ' in t is $\mathcal{I}^\star(t)^2/n + (\partial \Psi'/\partial E)(\partial E^\star/\partial t)$, where the first term is positive because $\omega(t) > 0$, and so is the second, because Ψ' strictly decreases in E (Lemma 3(d)) and $E^\star$ strictly decreases in t (Lemma 2(i)). Hence the loss strictly increases in t.

The loss vanishes in large samples. By Equation (2), $n(1 - \Psi') = \tau C'(\tau) = \underline{E}\tau + \int_0^t \mathcal{I}^\star(1 - \mathcal{I}^\star) \geq \underline{E}\tau$, and $n\Psi' = \int_0^t \mathcal{I}^{\star 2} \leq t$, so $M - 1 = \Psi'/(1 - \Psi') \leq t/(\underline{E}\tau)$. At noise ratio $\xi + \rho$, the floor is $\underline{E} = \xi + \rho + 1 - \Omega(t) \geq \xi + \rho$, and $E^\star_\rho \leq 1 + \xi + \rho$ gives $\tau_\rho \geq n/(1 + \xi + \rho)$. So $M_\rho - 1 \leq t(1 + \xi + \rho)/\big((\xi + \rho)n\big) \leq t(1 + \xi)/(\xi n)$, and integrating over $\rho \in [0, \nu]$ bounds the loss by $\nu\, t(1 + \xi)/(\xi n)$, which vanishes as $n \to \infty$.

The loss vanishes in small samples. Since $E^\star_\rho \geq \xi$, $\mathcal{I}^\star \leq n\bar{\omega}/\xi$, so $\Psi' \leq nt\bar{\omega}^2/\xi^2$. For $n \leq \xi^2/(2t\bar{\omega}^2)$, $\Psi' \leq 1/2$ and $M_\rho - 1 \leq 2nt\bar{\omega}^2/\xi^2$, so the loss is at most $2\nu nt\bar{\omega}^2/\xi^2$, which vanishes as $n \to 0$. □

B Simulations

I check how accurately the high-dimensional limit of Theorem 1 describes finite environments, and how the results change when the data depart from the Gaussian, independent specification of Section 2. The limit is within 1.1 percent of simulated value at every dimension simulated, the smallest being $K = 100$ (Appendix B.2). With non-Gaussian data, the value of the ridge predictor stays within 2.9 percent of the limit (Appendix B.3). The increasing returns to scale of Figure 5 appear already with $K = 200$ covariates and with non-Gaussian data (Appendix B.4). With correlated covariates, the patterns of the returns persist and value rises; the theory covers correlation only when the firm trains on and deploys every covariate, and in that case it predicts the gain (Appendix B.5).

B.1 Design and Replication

All exercises except the one on increasing returns to scale use the exponential profile of Figure 3, with decay rate $\delta = 0.5$, and $\xi = 0.2$. For a given K, a design $(N, T, T_0) = (nK, tK, t_0K)$ is simulated many times, and the captions report the number of replications. Each draw generates the weights, the covariates of N training individuals and the noise, and the firm estimates the weights of its T trained covariates by ridge regression with penalty $\bar{c} = \underline{E}(T/K)$, the misspecification floor at finite K. In the Gaussian model, each ridge coefficient differs from the

Table 3. Finite environments: simulated value against the limit

(n, t, t_0)	$\mathcal{V}(n, t, t_0)$	$K = 100$	$K = 400$	$K = 1600$
(0.25, 0.7, 0.7)	0.154	0.156 (0.003)	0.155 (0.002)	0.153 (0.002)
(1, 0.7, 0.7)	0.440	0.438 (0.003)	0.443 (0.002)	0.438 (0.001)
(4, 0.7, 0.7)	0.668	0.667 (0.001)	0.669 (0.001)	0.667 (0.000)
(1, 0.7, 0.35)	0.247	0.246 (0.002)	0.249 (0.001)	0.247 (0.001)
(1, 0.3, 0.3)	0.191	0.192 (0.002)	0.193 (0.001)	0.191 (0.001)
(4, 0.3, 0.15)	0.155	0.155 (0.001)	0.156 (0.000)	0.155 (0.000)

Gaussian data, exponential profile ($\delta = 0.5$), $\xi = 0.2$. Ridge predictor with penalty $\bar{c}$. Replications: 400, 200 and 100 for $K = 100$, 400 and 1600.

corresponding posterior mean by $O(K^{-1/2})$ in L^2, uniformly over trained covariates (Step 2 of the proof of Theorem 1), so the value of the ridge predictor is within $O(K^{-1})$ of the optimal value. In every model the ridge predictor is feasible, so its value bounds the value of the optimal predictor from below. The bound concerns levels; it carries no information about the curvature of optimal value. For each draw, the value of the predictor is $2\hat{\boldsymbol{\beta}}^\top \Sigma \boldsymbol{\beta} - \hat{\boldsymbol{\beta}}^\top \Sigma \hat{\boldsymbol{\beta}}$, where $\hat{\boldsymbol{\beta}}$ stacks the estimated weights of the T_0 deployed covariates, with zeros elsewhere, and Σ is the covariance matrix of the target individual's covariates; its expectation is $\mathrm{Var}(y_0) - \mathbb{E}[(y_0 - \hat{y})^2]$. Reported means use the control variate $\sum_{j,k \le T_0} \beta_j \Sigma_{jk} \beta_k - \sum_{k \le T_0} \Sigma_{kk}$, which has mean zero, and designs that differ only in n or t share their draws. Tables report standard errors in parentheses, and figures show 95 percent confidence intervals.

The folder `simulations` contains the Python code: `simcore.py` implements the simulation and the limit formulas, and one script per exercise writes the results from which `make_appendix.py` builds this appendix. Seeds are fixed, and the full set runs in about fifteen minutes on two cores. Setting the environment variable `REPS_MULT` multiplies every replication count, and `make_appendix.py` then updates every number in this appendix.

B.2 Finite Environments

The limit describes finite environments closely. Table 3 compares simulated values with the limit $\mathcal{V}(n, t, t_0)$ for $K \in \{100, 400, 1600\}$. Every difference lies within two standard errors. The largest absolute difference is 0.003 and the largest relative difference 1.1 percent; already at $K = 100$, the largest absolute difference is 0.002. The isotropic benchmark gives the same result, with a largest absolute difference of 0.003. Figure 7 turns to the objects behind the results. The ridge analogue of the fraction learned, $1 - \mathbb{E}[(\beta_k - \hat{\beta}_k)^2]$ for the ridge coefficient $\hat{\beta}_k$, which equals $\mathcal{I}_k(D)$ when $\hat{\beta}_k$ is the posterior mean, matches the single-weight value $\mathcal{I}(\omega(u); n, E^\star)$ of Theorem 1(ii) (panel a), and the marginal value of scope follows the U shape of Figure 4(a): it falls while decay dominates, reaches its minimum near $t = 0.45$, and rises once spillovers take over (panel b).

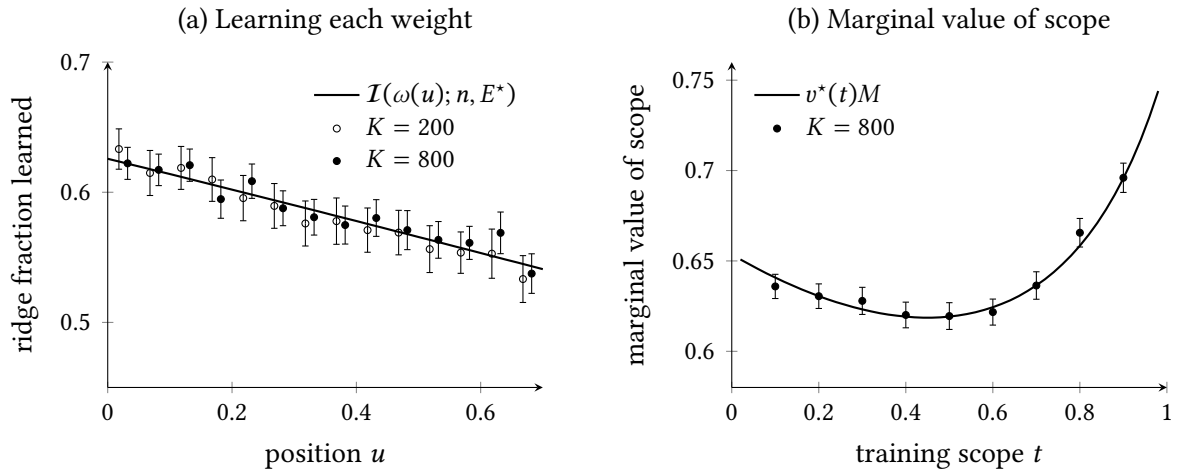


Figure 7. Finite environments. Gaussian data, exponential profile ($\delta = 0.5$), $\xi = 0.2$, $n = 1$, full deployment. (a) Ridge analogue of the fraction learned, $1 - \mathbb{E}[(\beta_k - \hat{\beta}_k)^2]$, by position at $t = 0.7$, averaged within bins of width 0.05. (b) Marginal value of scope, estimated by differences of simulated value over steps of 0.1 in t. Bars show 95 percent confidence intervals. Replications: 400 ($K = 200$) and 150 ($K = 800$) in (a), 600 in (b).

Table 4. Non-Gaussian data: simulated value against the limit, $K = 800$

(n, t, t_0)	$\mathcal{V}(n, t, t_0)$	Gaussian	Rademacher	Student t_5
(0.25, 0.7, 0.7)	0.154	0.155 (0.002)	0.156 (0.002)	0.150 (0.002)
(1, 0.7, 0.7)	0.440	0.441 (0.002)	0.442 (0.001)	0.438 (0.002)
(4, 0.7, 0.7)	0.668	0.668 (0.000)	0.667 (0.000)	0.667 (0.001)
(1, 0.7, 0.35)	0.247	0.248 (0.001)	0.248 (0.001)	0.246 (0.001)
(1, 0.3, 0.3)	0.191	0.192 (0.001)	0.192 (0.001)	0.190 (0.001)
(4, 0.3, 0.15)	0.155	0.155 (0.000)	0.155 (0.000)	0.155 (0.000)

Weights, covariates and noise share the distribution in each column, scaled to the variances of Section 2. Exponential profile ($\delta = 0.5$), $\xi = 0.2$, ridge predictor with penalty $\bar{c}$, 150 replications.

B.3 Non-Gaussian Data

The value of the ridge predictor stays close to the limit when the data are not Gaussian. Table 4 draws the weights, covariates and noise from a Rademacher distribution and from a Student t distribution with five degrees of freedom, scaled to preserve every second moment. The posterior mean is then nonlinear, and the ridge predictor is no longer optimal. The largest absolute and relative differences are 0.002 and 1.4 percent under the Rademacher distribution, and 0.004 and 2.9 percent under the heavy-tailed t distribution, both at the smallest scale. The optimal predictor does at least as well as the ridge predictor, so the limit is approximately a lower bound on the value of data. The universality of the Marčenko–Pastur and Silverstein–Bai laws (Bai and Silverstein 2010) points the same way, but spectral universality alone does not deliver the limit of the prediction risk, so I present these results as numerical support.

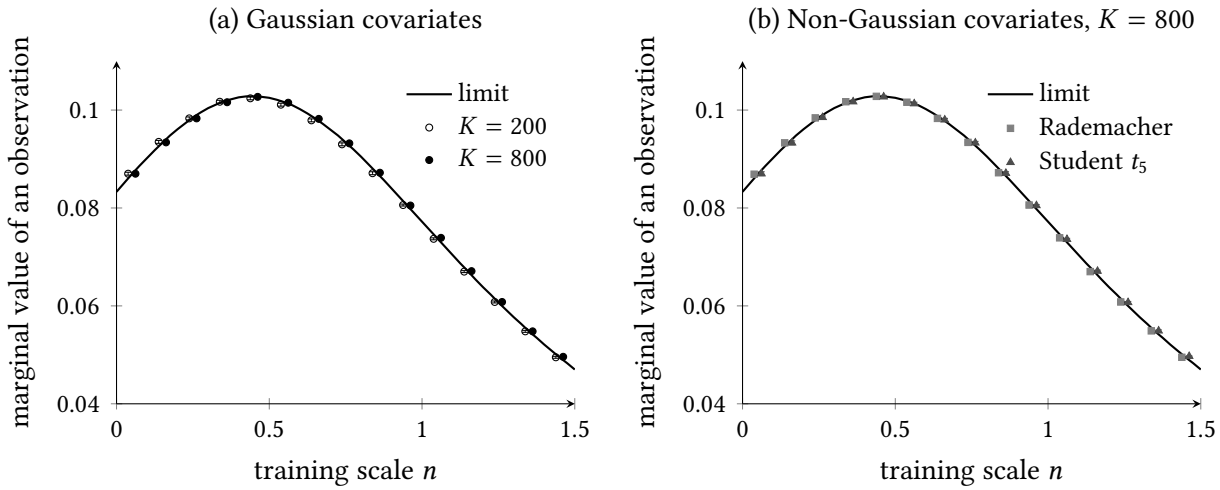


Figure 8. Increasing returns to scale at finite K. Example of Figure 5: loading 0.5 on the first 40 percent of covariates, 1.5 on the next 50 percent and 0.5 on the last 10 percent, $\xi = 0.2$, $t = 0.9$, $t_0 = 0.4$. Marginal value of an observation, estimated by differences of simulated value over steps of 0.1 in n; solid curve, the limit. Bars show 95 percent confidence intervals, narrower than the markers. Replications: 400 for $K = 200$ and 60 for $K = 800$.

B.4 Increasing Returns to Scale

The increasing returns to scale of Proposition 4 appear at small dimensions. Figure 8 simulates the example of Figure 5, in which the firm deploys covariates weaker than some it trains ($t = 0.9$, $t_0 = 0.4$). Conditional on the training covariates, the expected value of the ridge predictor has a closed form,

$$\frac{1}{K}\sum_{k\leq T_0}\omega_k\left[(\boldsymbol{A}+\bar{c}\,\boldsymbol{I})^{-1}\boldsymbol{A}\right]_{kk}, \qquad \boldsymbol{A}\equiv \boldsymbol{X}_{N,T}^{\top}\boldsymbol{X}_{N,T},$$

for any distribution of the weights, the errors and the untrained covariates with the second moments of Section 2. The exercise averages it over draws of the training covariates, nested across n, and estimates the marginal value of an observation by differences over steps of 0.1 in n. The marginal value rises up to $n \approx 0.45$ and falls beyond, as in the limit, already at $K = 200$, and with Rademacher and Student t_5 covariates at $K = 800$. The largest absolute difference from the limit is 0.0004, 0.4 percent of the marginal value.

B.5 Correlated Covariates

The returns of Section 4 are not an artifact of independence. Figure 9 and Table 5 correlate the covariates of each individual across positions, with $\text{corr}(x_{ik}, x_{ij}) = r^{|k-j|}$ for $r \in \{0.3, 0.6\}$, keeping their loadings; covariates adjacent in the order, which have similar strength, are the most correlated. Under full deployment, the marginal value of scope keeps its U shape, with its minimum at $t = 0.4$ or 0.5 for every r (Figure 9(a)), and the marginal value of scale falls with scale for every r (Figure 9(b)). Correlation raises the level of the marginal value of scope

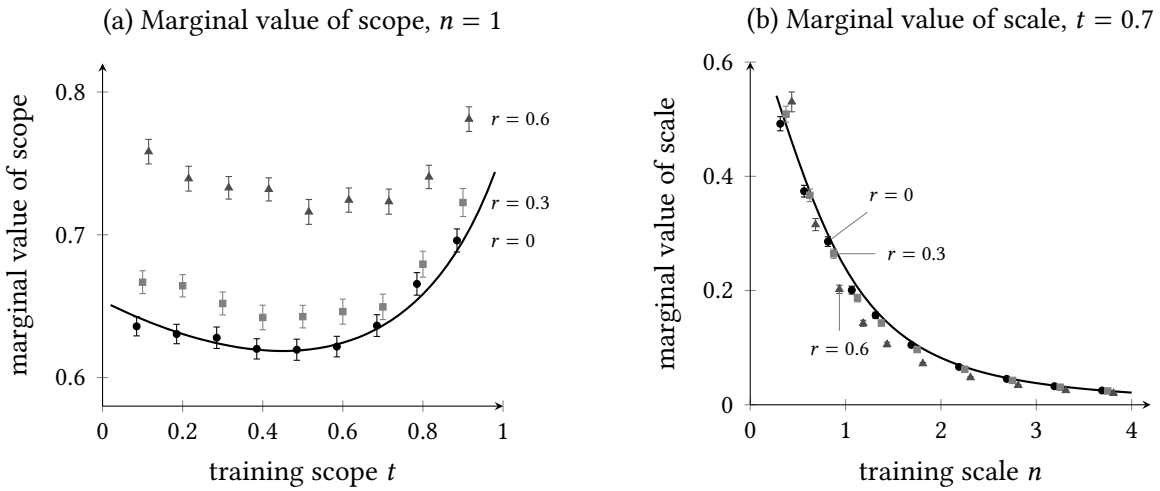


Figure 9. Correlated covariates, full deployment. Gaussian data, exponential profile ($\delta = 0.5$), $\xi = 0.2$, $K = 800$, $\text{corr}(x_{ik}, x_{ij}) = r^{|k-j|}$. Marginal values are differences of simulated value over steps of 0.1 in t (a) and between adjacent points of the grid of n (b); the solid curves are the limit for independent covariates. Bars show 95 percent confidence intervals; in (b), most are narrower than the markers. Replications: 600 for $r = 0$ and 450 for $r > 0$ in (a), 200 in (b).

and deepens its initial fall, consistent with collinearity making each new covariate partly redundant (Schäfer 2025). These patterns describe the ridge predictor, which differs from the posterior mean when correlated covariates are left untrained.

Correlation also raises value at every scope (Table 5). The theory covers correlation only when the firm trains on and deploys every covariate.

Two mechanisms can contribute. A trained covariate proxies for the correlated covariates the firm does not train, and correlation disperses the eigenvalues of the covariance matrix of the trained covariates, which makes their weights easier to learn from a given sample. Proxying is possible only with untrained covariates, whose signal share is only 0.04 at $t = 0.95$, yet a correlation of 0.6 raises value there from 0.61 to 0.70.

At $t = t_0 = 1$, no covariate is left untrained, so proxying is absent, the ridge predictor is the posterior mean, and the theory accounts for the gain. Rotating the covariates into the eigenbasis of their covariance matrix leaves the prior on the weights unchanged and turns the model into one with independent covariates whose loadings are the eigenvalues of $K\Sigma$. By the Schur–Horn theorem, the loadings ω_k, the diagonal of $K\Sigma$, are majorized by these eigenvalues (Marshall et al. 2011), so correlation spreads the loadings, and Proposition 9 implies that, in the limit, it raises value. The proof of Theorem 1 uses the loadings only through a uniform bound and the L^1 convergence of their step function, which sorted eigenvalues satisfy when their empirical distribution converges, with the quantile function of the limit as the profile. Theorem 1 applied to the eigenvalues predicts 0.669 and 0.744 for $r = 0.3$ and 0.6; the simulated values are 0.666 and 0.744, against 0.644 without correlation (150 replications). The gain at $t = 0.95$ is of the same size, so it too comes mainly from the spectrum of the trained covariates.

Table 5. Correlated covariates: value of full deployment, $n = 1$

	$t = 0.15$	$t = 0.35$	$t = 0.55$	$t = 0.75$	$t = 0.95$
Theory, $r = 0$	0.097	0.222	0.346	0.472	0.608
$r = 0$	0.096 (0.000)	0.222 (0.001)	0.346 (0.001)	0.472 (0.001)	0.608 (0.001)
$r = 0.3$	0.101 (0.000)	0.232 (0.001)	0.361 (0.001)	0.490 (0.001)	0.630 (0.001)
$r = 0.6$	0.115 (0.000)	0.262 (0.001)	0.407 (0.001)	0.552 (0.001)	0.704 (0.001)

Gaussian data, exponential profile ($\delta = 0.5$), $\xi = 0.2$, $K = 800$; 600 replications for $r = 0$ and 450 for $r > 0$.

C Additional Results

I extend the model in two directions and solve two special cases. Appendix C.1 lets the firm train on and deploy any set of covariates, not only the first ones in the order. Value keeps the form of Proposition 1, and in the limit the marginal value of a group of covariates rises with the covariates already trained. Sections 4.3 and 5 use these results for covariates added in random order and for holders of covariates. Appendix C.2 varies the primitives other than the design: ordering covariates by strength raises value, and so does spreading their loadings at a given signal share; noise lowers learning precision and the learning multiplier. Appendix C.3 solves two special cases, identical covariates and two covariates.

C.1 Arbitrary Sets of Covariates

The design of Section 2.2 trains on and deploys initial segments of one order. Here I drop that restriction, first at finite K and then in the limit.

Two orders. Suppose covariates become available for training in one order and for prediction in another. Let π and π' be permutations of $\{1, \dots, K\}$ that list the covariates in the training order and in the target order. A design (N, T, T_0) then trains on the set $\mathcal{T} \equiv \{\pi(1), \dots, \pi(T)\}$ and observes the set $\mathcal{P} \equiv \{\pi'(1), \dots, \pi'(T_0)\}$ for the target individual. For a set $\mathcal{S}$ of covariates, let $\mathcal{I}_k(N, \mathcal{S})$ be the fraction learned of covariate k's weight when the firm trains on $\mathcal{S}$ with N individuals.

Corollary 6 (Two orders)**.** *For every design* (N, T, T_0)*:*

(i) value depends on the design only through N*,* $\mathcal{T}$ *and* $\mathcal{T} \cap \mathcal{P}$*:*

$$V = \frac{1}{K} \sum_{k \in \mathcal{T} \cap \mathcal{P}} \omega_k \, \mathcal{I}_k(N, \mathcal{T});$$

(ii) relabel the covariates so that those in $\mathcal{T} \cap \mathcal{P}$ *come first, those in* $\mathcal{T} \setminus \mathcal{P}$ *next and the others last, each group in the training order. In the relabeled model, the single-order design* $\left(N, |\mathcal{T}|, |\mathcal{T} \cap \mathcal{P}|\right)$ *has the same value.*

Proof. *(i).* The proofs of Lemma 1 and Proposition 1 use the covariates observed only through the independence of the primitives, never through their order (see the proof of Proposition 6). Proposition 1 therefore holds with $\mathcal{T}$ in place of the first T covariates and $\mathcal{P}$ in place of the first T_0, which gives (i).

(ii). The covariates are independent across coordinates, and the prior on the weights treats them symmetrically, so relabeling them changes neither the joint law of the data that corresponding designs generate nor any fraction learned. The relabeled design trains on the same covariates and deploys the same ones, and (i) gives equal values. □

The relabeled order sorts covariates by use. Covariates the firm observes both in its training sample and when it predicts come first, and covariates available only for training come next. When a rule bars a covariate from use at prediction, the covariate moves from the first group to the second. By part (i), value falls by the covariate's deployed value, and the covariate's contribution to learning is unchanged. Proposition 4 carries over in the same terms: returns to observations can increase only if some covariate available only for training is stronger than some deployed covariate.

Any part of the covariates, in the limit. The results on covariates added in random order (Section 4.3) and on holders of covariates (Section 5) use the limit value of training on and deploying any part of the covariates. For a measurable $g : [0,1] \to [0,1]$, read $g(u)$ as the fraction of the covariates near position u that the firm trains on and deploys, and let

$$\Psi_g(E) \equiv 1 + \xi - \int_0^1 g(u)\, v\big(\omega(u); n, E\big)\, du, \qquad \Psi_g'(E) = \frac{1}{n}\int_0^1 g(u)\, \mathcal{I}\big(\omega(u); n, E\big)^2\, du.$$

Since $v(\omega; n, E) = n\omega^2/(n\omega + E)$ is decreasing and convex in E, Ψ_g is nondecreasing and concave, with $\Psi_g(0) = 1 + \xi - \int_0^1 g\,\omega \geq \xi$ and $\Psi_g \leq 1 + \xi$. It therefore has a unique fixed point $E^\star(g) \in [\xi, 1+\xi]$, and concavity gives $1 - \Psi_g'(E^\star(g)) \geq \Psi_g(0)/E^\star(g) \geq \xi/(1+\xi)$. Write

$$M(g) \equiv \frac{1}{1 - \Psi_g'\big(E^\star(g)\big)}, \qquad \mu(u; g) \equiv v\big(\omega(u); n, E^\star(g)\big)\, M(g),$$
$$\mathcal{V}(g) \equiv \int_0^1 g(u)\, v\big(\omega(u); n, E^\star(g)\big)\, du = 1 + \xi - E^\star(g).$$

For a set $S \subseteq [0,1]$ that is a finite union of intervals, $\mathcal{V}(\mathbf{1}_S)$ is the limit of the value of training on and deploying the covariates whose positions lie in S: relabel these covariates to come first (Corollary 6); Step 5 of the proof of Theorem 1 uses the loadings only through the L^1 convergence of their step function, which holds for the relabeled covariates.

Lemma 4 (Combining covariates in the limit)**.** *Let $g, g', h : [0,1] \to [0,1]$ be measurable with $g \leq g'$ and $g' + h \leq 1$. Then:*

(i) $\mathcal{V}(g+h) - \mathcal{V}(g) = \int_0^1 \int_0^1 h(u)\, \mu(u; g + sh)\, du\, ds$;

(ii) $E^\star(g') \le E^\star(g)$ *and* $M(g') \ge M(g)$*, so* $\mu(u; g') \ge \mu(u; g)$ *for every* u*;*

(iii) $\mathcal{V}(g' + h) - \mathcal{V}(g') \ge \mathcal{V}(g + h) - \mathcal{V}(g)$.

Proof. *(i).* The function $(E, s) \mapsto E - \Psi_{g+sh}(E)$ is continuously differentiable, with derivative $1 - \Psi'_{g+sh}(E) \ge \xi/(1+\xi)$ in E at the fixed point and $\int_0^1 h\, v(\omega; n, E)\, du$ in s. The implicit function theorem gives $dE^\star(g + sh)/ds = -M(g + sh) \int_0^1 h\, v\big(\omega; n, E^\star(g + sh)\big)\, du$, and $\mathcal{V} = 1 + \xi - E^\star$ gives (i).

(ii). Since $v \ge 0$, $\Psi_{g'} \le \Psi_g$, so $\Psi_{g'}(E^\star(g)) \le E^\star(g)$. A concave function $\Psi_{g'}$ with $\Psi_{g'}(0) > 0$ lies above the diagonal below its fixed point and below it above, so $E^\star(g') \le E^\star(g)$. Since $\mathcal{I}$ decreases in E and $g' \ge g$, $\Psi'_{g'}(E^\star(g')) \ge \Psi'_g(E^\star(g')) \ge \Psi'_g(E^\star(g))$, so $M(g') \ge M(g)$; and v decreases in E.

(iii). Apply (i) to both differences and (ii) to $g' + sh \ge g + sh$. □

Random order. Suppose the order is a uniformly random permutation of the covariates, drawn independently of everything else. The first T covariates are then a random sample, without replacement, of fraction T/K of the K covariates. In Step 5 of the proof of Theorem 1, each average $\frac{1}{K}\sum_{j\le T} f_{E,m}(\omega_j)$ has mean $\frac{T}{K}\cdot\frac{1}{K}\sum_{k\le K} f_{E,m}(\omega_k)$ and variance at most $\bar{\omega}^2/K$, and the functions $f_{E,m}$ are Lipschitz in (E, m) uniformly in ω, so the averages converge in probability to $t\int_0^1 f_{E,m}(\omega(u))\, du$, uniformly over $E \in [\xi, 1 + \xi]$. The rest of the proof goes through in probability: learning noise converges to $E^\star(t\mathbf{1}_{[0,1]})$, and value under full deployment to $\mathcal{V}(t\mathbf{1}_{[0,1]})$, the value of training on the fraction t of the covariates at every position. By Lemma 4(i) with $g = 0$ and $h = t\mathbf{1}_{[0,1]}$, $\mathcal{V}(t\mathbf{1}_{[0,1]}) = \int_0^t \int_0^1 \mu(u; s\mathbf{1}_{[0,1]})\, du\, ds$, so $d\mathcal{V}(t\mathbf{1}_{[0,1]})/dt = \int_0^1 \mu(u; t\mathbf{1}_{[0,1]})\, du$. Since $\int_0^1 \omega = 1$, Ψ_{t1} strictly decreases in t at every E, so $E^\star(t\mathbf{1}_{[0,1]})$ strictly decreases in t, and $\mu(u; t\mathbf{1}_{[0,1]})$ strictly increases in t wherever $\omega(u) > 0$. Value under full deployment is therefore strictly convex in t at every scale.

Holders of covariates. Let holders $i = 1, \dots, H$ own disjoint sets $S_i \subseteq [0, 1]$, each a finite union of intervals. When the holders in a coalition $J \subseteq \{1, \dots, H\}$ combine their covariates on the same individuals, and the firm trains on and deploys them, the limit value is $\Pi(J) \equiv \mathcal{V}(\mathbf{1}_{S_J})$, with $S_J \equiv \bigcup_{i\in J} S_i$.

Proposition 7 (Holders of covariates)**.** *For every profile and* $n > 0$*, the game* Π *is convex: for all* $J \subseteq J'$ *and* $i \notin J'$*,* $\Pi(J' \cup \{i\}) - \Pi(J') \ge \Pi(J \cup \{i\}) - \Pi(J)$*. Hence its Shapley value lies in its core.*

Proof. Apply Lemma 4(iii) with $g = \mathbf{1}_{S_J}$, $g' = \mathbf{1}_{S_{J'}}$ and $h = \mathbf{1}_{S_i}$. The Shapley value of a convex game lies in its core (Shapley 1971). □

C.2 The Profile and the Noise Ratio

Learning noise depends on the primitives other than the design through the map Ψ. I derive three comparative statics: in the order of the covariates, in the dispersion of their loadings, and in the noise ratio.

Order. Remark 3 states that a firm free to choose the order would take covariates in decreasing order of loading. For a profile ω, let $\omega^{\downarrow}$ be its nonincreasing rearrangement, $\omega^{\downarrow}(s) \equiv \inf\left\{\lambda \geq 0 : |\{u \in [0,1] : \omega(u) > \lambda\}| \leq s\right\}$, and write $\Omega^{\downarrow}$, $\Psi^{\downarrow}$, $E^{\star\downarrow}$ and $\mathcal{V}^{\downarrow}$ for the signal share, fixed-point map, learning noise and limiting value it generates. The rearrangement is nonincreasing and equimeasurable with ω: the two functions have the same distribution under Lebesgue measure, $|\{u : \omega(u) \in B\}| = |\{u : \omega^{\downarrow}(u) \in B\}|$ for every Borel set B (Hardy et al. 1952, ch. 10). It is therefore itself a profile, with values in $[0, \bar{\omega}]$ and integral one. Since σ_β^2 multiplies both values, take $\sigma_\beta^2 = 1$.

Proposition 8 (Strongest first is optimal)**.** *For every profile ω, $n > 0$ and $0 \leq t_0 \leq t \leq 1$:*

(i) $\Omega(t) \leq \Omega^{\downarrow}(t)$;

(ii) $E^{\star\downarrow}(n,t) \leq E^{\star}(n,t)$;

(iii) $\mathcal{V}(n,t,t_0) \leq \mathcal{V}^{\downarrow}(n,t,t_0)$.

Proof. First, for every nondecreasing $\chi : [0,\bar{\omega}] \to [0,\infty)$ and every $s \in [0,1]$,

$$\int_0^s \chi\big(\omega(u)\big)\,du \leq \int_0^s \chi\big(\omega^{\downarrow}(u)\big)\,du. \tag{13}$$

By the layer-cake formula, $\int_0^s \chi(f(u))\,du = \int_0^\infty |\{u \leq s : \chi(f(u)) > \lambda\}|\,d\lambda$ for $f \in \{\omega, \omega^{\downarrow}\}$. For each λ, $U_\lambda \equiv \{x : \chi(x) > \lambda\}$ is an upper interval of $[0,\bar{\omega}]$, because χ is nondecreasing. Equimeasurability gives $|\{u : \omega(u) \in U_\lambda\}| = |\{u : \omega^{\downarrow}(u) \in U_\lambda\}| \equiv \ell_\lambda$, and the second set is an initial interval of $[0,1]$, because $\omega^{\downarrow}$ is nonincreasing. Hence $|\{u \leq s : \omega(u) \in U_\lambda\}| \leq \min\{s, \ell_\lambda\} = |\{u \leq s : \omega^{\downarrow}(u) \in U_\lambda\}|$, and integrating over λ gives (13).

Part (i) is (13) with $\chi(x) = x$. For (ii), fix $E > 0$. The map $x \mapsto v(x;n,E) = nx^2/(nx+E)$ is nondecreasing and nonnegative, so (13) gives $\Psi^{\downarrow}(E;n,t) \leq \Psi(E;n,t)$ for every E. At $E = E^{\star}(n,t)$ this reads $E^{\star} \geq \Psi^{\downarrow}(E^{\star};n,t)$, and the crossing property (6) (Step 1 of the proof of Theorem 1), applied to $\omega^{\downarrow}$, gives $E^{\star} \geq E^{\star\downarrow}$. For (iii), (13) and the fact that v is decreasing in E give

$$\mathcal{V}(n,t,t_0) = \int_0^{t_0} v\big(\omega;n,E^{\star}\big)\,du \leq \int_0^{t_0} v\big(\omega^{\downarrow};n,E^{\star}\big)\,du \leq \int_0^{t_0} v\big(\omega^{\downarrow};n,E^{\star\downarrow}\big)\,du = \mathcal{V}^{\downarrow}(n,t,t_0).$$

□

Dispersion. At a given signal share, the dispersion of the loadings also matters. Say that a profile $\tilde{\omega}$ is *more dispersed* than ω on $[0,t]$ if $\int_0^t \chi(\omega(u))\,du \le \int_0^t \chi(\tilde{\omega}(u))\,du$ for every convex $\chi : [0,\infty) \to \mathbb{R}$, and write $\tilde{\Psi}$, $\tilde{E}^\star$ and $\tilde{\mathcal{V}}$ for the fixed-point map, learning noise and limiting value that $\tilde{\omega}$ generates. Taking $\chi(x) = x$ and $\chi(x) = -x$ shows that the two profiles then carry the same signal share $\Omega(t)$.

Proposition 9 (Dispersion)**.** *Let $\tilde{\omega}$ be more dispersed than ω on $[0,t]$, with $t \in (0,1]$ and $n > 0$. Then $\tilde{E}^\star(n,t) \le E^\star(n,t)$ and $\tilde{\mathcal{V}}(n,t,t) \ge \mathcal{V}(n,t,t)$. In particular, among profiles with signal share $\Omega(t)$, the profile that is constant on $[0,t]$ minimizes $\mathcal{V}(n,t,t)$.*

Proof. For fixed n and $E > 0$, the map $x \mapsto v(x;n,E) = nx^2/(nx+E)$ is convex on $[0,\infty)$, with second derivative $2nE^2/(nx+E)^3 > 0$. Hence $\tilde{\Psi}(E;n,t) \le \Psi(E;n,t)$ for every E. At $E = E^\star$ this reads $E^\star \ge \tilde{\Psi}(E^\star;n,t)$, and the crossing property (6), applied to $\tilde{\omega}$, gives $E^\star \ge \tilde{E}^\star$. Since value under full deployment equals $1+\xi$ minus learning noise for every profile (Corollary 2), $\tilde{\mathcal{V}}(n,t,t) \ge \mathcal{V}(n,t,t)$. For the last claim, Jensen's inequality shows that every profile with signal share $\Omega(t)$ is more dispersed on $[0,t]$ than the profile equal to $\Omega(t)/t$ there. □

Predictive content concentrated in a few strong covariates is cheaper to learn, and a profile that is flat on the trained covariates is the hardest to learn from. Appendix B uses Proposition 9 to account for the value of correlated covariates.

Noise. The noise ratio $\xi = 1/R^2_{\max} - 1$ (Section 3.3) measures how predictable the target variable is at best. It enters Ψ additively, which gives the following comparative statics, used in Corollary 5.

Lemma 5 (Noise)**.** *Fix $n > 0$ and $t \in (0,1)$. Then $\partial E^\star/\partial\xi = M > 0$, and learning precision τ strictly decreases in ξ. If $\Omega(t) > 0$, the learning multiplier M strictly decreases in ξ.*

Proof. The noise ratio enters Ψ additively, $\partial\Psi/\partial\xi = 1$, so the implicit function theorem gives $\partial E^\star/\partial\xi = M$, and $\tau = n/E^\star$ falls. The slope $\Psi'(E;n,t) = \frac{1}{n}\int_0^t \mathcal{I}(\omega(u);n,E)^2\,du$ does not depend on ξ and, since $\omega > 0$ on a subset of $[0,t]$ of positive measure, strictly decreases in E by Lemma 3(d), so M falls. □

By Equation (2), at given scope, raising the noise ratio from ξ_0 to ξ raises the number of observations the firm needs to attain precision τ by $(\xi - \xi_0)\tau$. With a higher noise ratio, learning precision and the learning multiplier are lower, the gap in deployed value between weak and strong covariates is wider (Section 4.1), and learning noise is higher, so the half-learned condition of Proposition 5(ii) holds up to larger scales. In Figure 4(b), at a higher noise ratio the spillover rate is lower and peaks at a larger scale. Inequality (4), the condition for increasing returns to scale at small scale, bounds the noise ratio by a function of the loadings alone.

C.3 Two Special Cases

Identical covariates. When $\omega \equiv 1$, Equation (1) is quadratic in the fraction learned, and value has a closed form.

Corollary 7 (Isotropic benchmark). *If $\omega \equiv 1$, then $\Omega(t) = t$ and $\mathcal{V}(n, t, t_0) = t_0\, \mathcal{I}(n, t)$, where, with slight abuse of notation, $\mathcal{I}(n, t) \equiv \mathcal{I}\big(1; n, E^\star(n, t)\big)$ is the root in $[0, 1]$ of $t\mathcal{I}^2 - (n + 1 + \xi)\mathcal{I} + n = 0$:*

$$\mathcal{I}(n, t) = \frac{2n}{n + 1 + \xi + \sqrt{(n + 1 + \xi)^2 - 4nt}}.$$

Proof. With $\omega \equiv 1$, the equation $E = \Psi(E; n, t)$ reads $E = \xi + 1 - t + t(1 - \mathcal{I}) = \xi + 1 - t\mathcal{I}$, while $\mathcal{I} = n/(n + E)$ gives $E = n(1 - \mathcal{I})/\mathcal{I}$. Equating the two yields the quadratic; its smaller root, written in rationalized form, is the one in $[0, 1]$. □

In this benchmark, returns to target scope are constant (Proposition 2), returns to scope and returns to training scope alone increase at every scale (Corollary 4 and Proposition 3(iii)), and returns to scale diminish (Proposition 4(i)). The closed form also gives value in large samples. Writing $\mathcal{I} = 1 - \kappa/n + O(n^{-2})$ in the quadratic and matching the terms of order one gives $\kappa = 1 + \xi - t$, so

$$\bar{\mathcal{V}}(n, t) = t - \frac{t\,(1 + \xi - t)}{n} + O(n^{-2}), \qquad \frac{\partial \bar{\mathcal{V}}}{\partial n} = \frac{t\,(1 + \xi - t)}{n^2} + O(n^{-3}),$$
$$\frac{\partial^2 \bar{\mathcal{V}}}{\partial n\, \partial t} = \frac{1 + \xi - 2t}{n^2} + O(n^{-3}),$$

where the expansion can be differentiated term by term because $\mathcal{I}$ is analytic in $(1/n, t)$ near $1/n = 0$. Equivalently, $E^\star = \underline{E}(t)(1 + t/n) + O(n^{-2})$: to first order, the error of least squares on T covariates with N observations, the floor inflated by $N/(N - T)$. Under full deployment, scale and scope therefore remain complements in large samples when $t < (1 + \xi)/2$, and become substitutes when $t > (1 + \xi)/2$.

Two covariates. The mechanism of Section 4.4 holds with two covariates and without idiosyncratic noise. Let $K = 2$ and $\xi = 0$, so that the loadings satisfy $\omega_1 + \omega_2 = 2$, and let the firm train on both covariates and deploy the first. Proposition 1 applies at fixed K. Without data, value is zero. One observation, $y_1 = \beta_1 x_{11} + \beta_2 x_{12}$, carries information about both weights, and the firm learns the fraction $\sqrt{\omega_1}/(\sqrt{\omega_1} + \sqrt{\omega_2})$ of the first. Two observations reveal both weights. By Proposition 1(iii), the first observation is worth $\frac{\omega_1}{2}\sqrt{\omega_1}/(\sqrt{\omega_1} + \sqrt{\omega_2})$ and the second $\frac{\omega_1}{2}\sqrt{\omega_2}/(\sqrt{\omega_1} + \sqrt{\omega_2})$. The second observation is worth more than the first if and only if $\omega_2 > \omega_1$, that is, if and only if the covariate the firm trains on without deploying is the stronger one: from the first observation the firm learns mostly the strong weight, and from the second the weak weight it deploys. If the firm deploys both covariates, the first observation is worth $(\sqrt{\omega_1} - \sqrt{\omega_2})^2/2$ more than the second, so it is worth at least as much whatever the loadings.

D Connections to Statistics

Appendix D.1 relates the learning-noise representation to random matrix theory, and Appendix D.2 shows that approximate message passing attains learning noise.

D.1 Relation to Random Matrix Theory

Equation (1) is the Silverstein–Bai equation for the limiting spectrum of a sample covariance matrix with population covariance $\mathrm{diag}(\omega_k)_{k\le T}$ (Silverstein and Bai 1995; Bai and Silverstein 2010): $1/E^\star(n,t)$ is the limit of the expected Stieltjes transform of the empirical spectral distribution of $\boldsymbol{X}_{N,T}\boldsymbol{X}_{N,T}^\top$ at $-\underline{E}(t)$. In the isotropic benchmark, the equation reduces to the Marčenko–Pastur law (Marčenko and Pastur 1967). These results deliver learning noise, a normalized trace. Value also depends on the fraction learned of each deployed covariate, which depends on the eigenvectors of the sample covariance matrix; random matrix theory obtains such quantities from deterministic equivalents of its resolvent (Hachem et al. 2007; Couillet and Liao 2022). Bordelon et al. (2020) obtain both objects for kernel regression by replica calculations, which are not rigorous, and find $1/(1-\Psi')$, the learning multiplier of Section 4.2, as a factor of the prediction error. The representation is also an instance of the decoupling principle of statistical physics, under which each unknown of a large linear system is seen through a scalar Gaussian channel whose noise, learning noise here, solves a fixed point (Guo and Verdú 2005). The proof of Theorem 1 derives these objects directly. It treats the untrained covariates as extra Gaussian noise, computes the fraction learned of one weight at a time, and ties these fractions together through an exact finite-sample identity, Equation (9) in Appendix A. By this route, I obtain the convergence of every fraction learned, uniformly across covariates, allow the firm to deploy fewer covariates than it trains on, and obtain the stability condition $\Psi'(E^\star;n,t)<1$ on which the comparative statics of Section 4 rest.

Degrees of freedom. The learned scope of Section 4.4 counts degrees of freedom: $K\tilde{t}\approx\sum_{k\le T}\mathcal{I}_k(D)$ is the effective number of parameters of ridge regression, the trace of the smoother matrix (Hastie et al. 2009), and Equation (2) is the limit of the exact identity in Equation (9); Simon et al. (2023) obtain its analogue for kernel regression, in which the fractions learned sum to the sample size up to a regularization term. A firm that estimates the weights by least squares spends a full degree of freedom on each: its cost of learning the weights is t, it needs T observations to determine T weights, and its out-of-sample error is the floor inflated by $N/(N-T)$ (Hastie et al. 2022). This error diverges as N falls to T, which produces the peak of the double-descent curve (Belkin et al. 2019). By Equation (2), the error of the optimal predictor is the floor inflated by $N/(N-K\tilde{t})$; since $K\tilde{t}<N$ at every sample size, the inflation never diverges, and since $E^\star$ falls with N (Lemma 2(i)), the error has no peak.

D.2 Approximate Message Passing

Remark 2 states that approximate message passing attains learning noise. Here I define the algorithm, derive its state evolution and show that its fixed point is the ridge predictor of Section 3.3.

The algorithm. For $k \le T$, rescale each trained covariate and its weight, $z_{ik} \equiv x_{ik}/\sqrt{n\omega_k}$ and $\theta_k \equiv \sqrt{n\omega_k}\,\beta_k$. Then $\boldsymbol{y}_N = \boldsymbol{Z}\boldsymbol{\theta} + \boldsymbol{w}$, where $\boldsymbol{Z} \in \mathbb{R}^{N\times T}$ has i.i.d. $\mathcal{N}(0, 1/N)$ entries, $\theta_k \sim \mathcal{N}(0, n\omega_k)$, and $\boldsymbol{w}$ collects the untrained covariates and the idiosyncratic errors. Starting from $\boldsymbol{\theta}^0 = \mathbf{0}$, $\boldsymbol{r}^{-1} = \mathbf{0}$ and $b_0 = 0$, approximate message passing iterates, for $\ell = 0, 1, \dots,$

$$\boldsymbol{r}^\ell = \boldsymbol{y}_N - \boldsymbol{Z}\boldsymbol{\theta}^\ell + b_\ell\,\boldsymbol{r}^{\ell-1}, \qquad E_\ell = \tfrac{1}{N}\|\boldsymbol{r}^\ell\|^2,$$
$$\theta_k^{\ell+1} = \mathcal{I}(\omega_k; n, E_\ell)\left(\theta_k^\ell + (\boldsymbol{Z}^\top \boldsymbol{r}^\ell)_k\right), \qquad b_{\ell+1} = \tfrac{1}{N}\textstyle\sum_{k\le T}\mathcal{I}(\omega_k; n, E_\ell).$$

The vector $\boldsymbol{\theta}^\ell + \boldsymbol{Z}^\top\boldsymbol{r}^\ell$ behaves as $\boldsymbol{\theta}$ plus Gaussian noise of variance E_ℓ, and the update applies to each coordinate the posterior mean of one weight observed with noise of that variance. The term $b_\ell \boldsymbol{r}^{\ell-1}$ is the Onsager correction. Its coefficient, the average derivative of the update, is the effective number of parameters per observation (Bayati and Montanari 2012): in the limit, the learned scope $\tilde{t}$ of Equation (2) at precision $n/E_{\ell-1}$, divided by n. Each iteration thus adds back the part of the residual that the current estimates have fit. Without the correction, the iteration is iterative shrinkage, and its noise does not follow Ψ.

State evolution and optimality. State evolution gives, for each ℓ, $E_\ell \to E_\ell^\infty$ as $K \to \infty$, where $E_0^\infty = 1 + \xi$ and $E_{\ell+1}^\infty$ is the noise variance $\underline{E}(t)$ plus $\frac{1}{N}$ times the sum of the error variances of the updates,

$$E_{\ell+1}^\infty = \underline{E}(t) + \frac{1}{K}\sum_{k\le T}\frac{\omega_k E_\ell^\infty}{n\omega_k + E_\ell^\infty} \longrightarrow \Psi(E_\ell^\infty; n, t).$$

The rescaled design has i.i.d. Gaussian entries and independent Gaussian priors with covariate-specific variances, for which Berthier et al. (2020) prove this state evolution; Bayati and Montanari (2011) treat identical priors. The untrained covariates add Gaussian noise, as in Step 2 of the proof of Theorem 1. Since $\Psi(\cdot; n, t)$ is nondecreasing and has a unique fixed point, the iterates fall monotonically to learning noise, and the value of the algorithm's predictions converges to $\mathcal{V}(n, t, t_0)$. Started from the old learning noise after training scope rises, the iterations trace the staircase of Figure 3(a), whose steps are, approximately, the terms of the learning multiplier. For identical priors, the optimality of message passing when its fixed point is unique follows from Reeves and Pfister (2019) and Barbier et al. (2020); the result here also covers covariate-specific priors and designs that deploy fewer covariates than they train on.

The fixed point. At a fixed point $(\boldsymbol{\theta}, \boldsymbol{r}, E, b)$ of the iteration, the first update gives $\boldsymbol{r} = (\boldsymbol{y}_N - \boldsymbol{Z}\boldsymbol{\theta})/(1-b)$, and the second becomes, in terms of the weights $m_k = \theta_k/\sqrt{n\omega_k}$, the ridge normal equations

$$\left(\boldsymbol{X}_{N,T}^\top \boldsymbol{X}_{N,T} + \lambda \boldsymbol{I}\right)\boldsymbol{m} = \boldsymbol{X}_{N,T}^\top \boldsymbol{y}_N, \qquad \lambda = E(1-b).$$

This is the calibration that Bayati and Montanari (2012) derive for the LASSO, with a linear update in place of soft thresholding. Since $\omega(1-\boldsymbol{\mathcal{I}}) = (E/n)\,\boldsymbol{\mathcal{I}}$, the fixed-point equation $E = \Psi(E; n, t)$ reads $E = \underline{E}(t) + Eb$, so $\lambda \to \underline{E}(t)$: the algorithm converges to ridge regression with penalty equal to the misspecification floor, the predictor of Section 3.3.

Numerical check. In simulations with the profile of Figure 3 ($n = 1$, $t = 0.7$, $\xi = 0.2$, $K = 2000$), E_ℓ tracks the iterates of Ψ from $1+\xi$ to $E^\star(n,t)$, the fixed point coincides with ridge regression with penalty $E(1-b)$ to machine precision, and without the correction the noise departs from Ψ after the first iteration.